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Automated Detection and Morphological Classification of Mast Cells in Bone Marrow Aspirates Using Deep Learning

Bachelor Thesis

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Abstract

English Version

Systemic mastocytosis (SM) is diagnosed in part by a WHO minor criterion requiring at least ≥25% of mast cells in bone marrow aspirates to be morphologically atypical. Manual cell classification is slow, subjective, and error-prone. To our knowledge, no validated automated pipeline exists for the specific task of distinguishing atypical from normal mast cells in alignment with this criterion. This thesis develops and evaluates a deep learning pipeline that detects mast cells in 512×512 bone marrow aspirate tiles and classifies each as atypical or normal, reporting the fraction of atypical cells directly against the diagnostic threshold. A YOLO11n detector was trained on a dataset from 53 patients assembled through a bootstrapped, model-assisted annotation loop in which each model generation labelled the next patient batch and its confirmed false positives became the following generation's hard negatives. Patient-held-out cross-validation was applied throughout: a grouped 5-fold protocol across 35 annotated patients served as the primary generalization estimate (mAP50-95 = 0.748, recall on atypical = 0.885, recall on normal = 0.821; best multi-patient fold recall on normal = 0.981), and a 35-fold leave-one-patient-out run provided an optimistic upper bound (Recall on atypical = 0.920, recall on normal = 0.941). A controlled design-of-experiment showed that corpus expansion improves rare-class recall more than hyperparameter tuning under structural class imbalance. The final model is deployed at USZ as MastCellDetector, an offline desktop application that reports the fraction of atypical cells against the WHO threshold, accompanied by an out-of-sample confidence calibration reference for clinical threshold selection.

German Version

Die systemische Mastozytose (SM) wird unter anderem über ein WHO-Nebenkriterium diagnostiziert, das verlangt, dass mindestens 25 % der Mastzellen im Knochenmarkaspirat morphologisch atypisch sind. Die händische Zellklassifikation ist langsam, subjektiv und fehleranfällig. Nach unserem Kenntnisstand existiert für die spezifische Aufgabe, atypische von normalen Mastzellen im Sinne dieses Kriteriums zu unterscheiden, bislang keine validierte automatisierte Pipeline. Die vorliegende Arbeit entwickelt und evaluiert eine Deep-Learning-Pipeline, die Mastzellen in 512×512 grossen Kacheln von Knochenmarkaspiraten detektiert, sie jeweils als Atypisch oder Normal einstuft und den Anteil atypischer Zellen unmittelbar dem diagnostischen Schwellenwert gegenüberstellt. Ein YOLO11n-Detektor wurde auf einem Datensatz von 53 Patienten trainiert, der über eine bootstrap-artige, modellgestützte Annotationsschleife aufgebaut wurde: Jede Modellgeneration annotierte den jeweils nächsten Patienten-Batch, und deren bestätigte Falsch-Positive gingen als Hintergrundbilder in die Folgegeneration ein. Durchgängig kam eine Kreuzvalidierung mit konsequenter Patiententrennung zum Einsatz. Als primäre Schätzung der Generalisierungsleistung diente eine gruppierte 5-fache Kreuzvalidierung über 35 annotierte Patienten (mAP50-95 = 0,748; Recall atypisch = 0,885; Recall normal = 0,821; bester Multi-Patienten-Split Recall normal = 0,981). Ein 35-facher Leave-one-patient-out-Durchlauf lieferte zusätzlich eine optimistische Obergrenze (Recall atypisch = 0,920; Recall normal = 0,941). Eine statistische Versuchsplanung (DoE) zeigte, dass unter strukturellem Klassenungleichgewicht eine Erweiterung des Datenbestands den Recall der seltenen Klasse stärker verbessert als ein Hyperparameter-Tuning. Das finale Modell wird am USZ als MastCellDetector eingesetzt, einer offline lauffähigen Desktop-Anwendung, die den atypischen Anteil gegen den WHO-Schwellenwert ausweist und durch eine Out-of-Sample-Konfidenzkalibrierung als Referenz für die klinische Schwellenwahl ergänzt wird.

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List of Abbreviations

- **AI**: Artificial Intelligence
- **AP**: Average Precision
- **CNN**: Convolutional Neural Network
- **COCO**: Common Objects in Context
- **CV**: Cross Validation
- **CVAT**: Computer Vision Annotation Tool
- **DoE**: Design of Experiment
- **FN**: False Negative
- **FP**: False Positive
- **GT**: Ground Truth
- **IoU**: Intersection over Union
- **LOPO**: Leave-One-Patient-Out
- **MC**: Mast Cells
- **mAP**: Mean Average Precision
- **SM**: Systemic Mastocytosis
- **TN**: True Negative
- **TP**: True Positive
- **USZ**: University Hospital Zurich (Universitätsspital Zürich)
- **WHO**: World Health Organization
- **YOLO**: You Only Look Once

1. Introduction

1.1. Context and Motivation

Systemic mastocytosis (SM) is a rare clonal hematopoietic neoplasm defined by the pathological expansion and accumulation of neoplastic mast cells (MC) in extracutaneous organs, mostly seen in the bone marrow, but also the spleen, liver, and gastrointestinal tract. The major diagnostic criterion for SM is the presence of multifocal MC clusters in the bone marrow and/or extracutaneous organs, while minor criteria include an elevated serum tryptase level, MC expression of Cluster of Differentiation 2 (CD2), CD25, or CD30, and the presence of activating tyrosine kinase (KIT) mutations [1]. Despite these structured criteria, SM is frequently underdiagnosed in routine clinical practice [2]. One of the central reasons for this is that MC, although morphologically distinct, occur only in very small numbers within a given bone marrow aspirate sample, making their manual identification and quantification an exhausting, subjective, and time-consuming task for hematopathologists.

The morphological assessment of MC carries direct diagnostic weight. The abnormal morphology of MC, defined as atypical spindle-shaped cells with hypogranulated cytoplasm and an oval nucleus, represents the first minor SM criterion. At least ≥25% of all MC must exhibit these morphological features in bone marrow smears or sections for this criterion to be fulfilled [1]. This quantitative threshold requires a reliable and reproducible count of both typical and atypical MC within a sample, a task that is very error-prone when performed manually and at scale.

In clinical practice, the integration of clinical signs and symptoms together with bone marrow morphological, immunophenotypic, and molecular results is required to diagnose SM variants [3]. However, the morphological step remains the entry point into this diagnostic chain. Standardizing and automating this step through computational means therefore has direct implications for diagnostic accuracy and clinical throughput. The University Hospital Zurich (USZ), which operates as the clinical partner for this thesis, has a concrete need for such a tool that could be embedded in its existing cell classification infrastructure.

1.2. Research Questions

The goal of this thesis is to develop and evaluate a pipeline for the automated detection and morphological classification of MC in bone marrow aspirate images. From this objective, the following research questions are derived:

1. Can an DL-based pipeline reliably distinguish between typical and atypical mast cell phenotypes to support the ≥25% diagnostic criterion defined by the WHO 2022 classification [1]?
2. Which specific performance metrics (e.g., F1-score, Mean Average Precision, or false negative rate) are most critical to ensure the tool's reliability for clinical screening?
3. Is the developed pipeline capable of identifying mast cells in samples with extremely low cell density ("rare event detection") without generating an unmanageable number of false positives?

1.3. Overall Concept

The overall system was designed as a complete multi-stage pipeline including data acquisition, annotation, preprocessing, hyperparameter tuning, model training, model validation, and clinical deployment. Digital bone marrow aspirate images were collected and annotated by clinical experts at the USZ to produce a labeled training dataset. Initial hyperparameter tuning was applied to ensure optimal starting conditions for training. A stratified k-fold cross-validation strategy was utilized to deliver a robust model evaluation given the limited availability of annotated rare-cell data. A YOLO-family object detection model was selected and fine-tuned for this task. The final pipeline is intended to be accessible either via a user-friendly drag-and-drop interface that automatically presents detection results and a statistical summary of the analyzed sample, meeting the operational requirements set by the USZ or, where feasible, integrated into Cinderella, the cell classification software already in use at the USZ.

1.4. State of the Art

1.4.1. Deep Learning in Digital Hematology

The intersection of deep learning and hematological image analysis has produced noteworthy progress over the past years. Traditionally, the detection and classification of blood cells involved manual microscopic examination by pathologists, a process that is labor-intensive, subjective, and error-prone. The increasing range of diagnostic tests has driven demand for automated, high-throughput solutions that are cost-effective, require a rapid turnaround time, and provide reliable results. Convolutional neural networks (CNNs) and, more recently, transformer-based architectures have addressed this demand by learning to identify and categorize morphological features directly from raw pixel data [4].

Among the detection frameworks applied in this domain, the YOLO (You Only Look Once) family of models has proven particularly effective. Studies applying YOLOv10 and YOLOv11 to the detection and classification of red blood cells, white blood cells, and platelets in microscopic images have demonstrated mAP scores of up to 93.8%, underscoring the potential of these architectures in supporting hematological analysis and clinical diagnosis [5]. A feature of YOLO-based pipelines is their ability to perform detection and classification in a single forward pass, hence their name "… Look Once", making them computationally suitable for integration into clinical workflows where inference speed matters. For further information about the mechanisms for YOLO-models see [6].

A meta-analysis of over 400 peer-reviewed studies on deep learning in peripheral blood smear analysis confirmed significant advances in automated diagnostic tools, particularly for disease identification in blood cells, while also identifying critical gaps including the limited exploration of differential white blood cell counting, the underdevelopment of automated morphological analysis, and the absence of standardized evaluation criteria [7]. These gaps are directly relevant to the present thesis, which addresses an underexplored subtype of rare hematological cell analysis.

1.4.2. Comparable Work in Cell Detection

A directly relevant contribution in this domain was published by Glüge et al. [8] in collaboration with the USZ. The study evaluated four state-of-the-art CNN architectures on a dataset of 171′374 microscopic single-cell images from bone marrow smears of 945 patients, finding that the best pre-trained model achieved a 53.5% improvement in precision and 7.3% improvement in recall compared to training from scratch, and that out-of-domain pre-training yielded features that transferred well to cell classification tasks.

A review by Ghete et al. [9] covering DL-based cell classification in bone marrow aspirate smears from 2019 to 2024 provides a systematic overview of the broader field. The review found that performance metrics across published studies are *not directly comparable*, as the majority implemented their own methods on their own datasets, and few were validated on external datasets. This observation directly informed the design choices of the present thesis, which applies stratified cross-validation to ensure statistically robust performance estimates rather than relying on single-split evaluations.

For multi-class white blood cell classification at the subtype level, Diaz et al. [10] applied a YOLOv11-large model with spatial attention to the simultaneous localization and classification of 13 white blood cell subtypes. The model achieved a mAP50-95 of 77.9% with an overall F1-score of 0.913, with class-wise average precision ranging between 89.3% and 98.2%, confirming reliable recognition even for infrequent cell subtypes. This is particularly instructive for the present work, as the challenge of infrequent subtypes, low absolute count, high inter-class visual similarity, maps directly onto the atypical versus normal MC classification problem.

Finally, Lv et al. [11] reported on the Morphogo system, a CNN-based platform trained on over 2.8 million bone marrow cell images. Evaluated across 508 cases, the system demonstrated the ability to identify over 25 different cell types, achieving a sensitivity of 80.95%, a specificity of 99.48%, and an overall accuracy of 99.01%. The dataset size required to achieve these results underscores the data scarcity challenge that constrains MC-specific models, a bottleneck that the present thesis addresses through targeted augmentation and cross-validation strategies.

1.4.3. Mast Cell Detection and the Clinical Gap

In contrast to blood cell detection, which benefits from large, publicly available datasets, automated MC analysis in bone marrow aspirates remains largely unexplored in the literature. As shown in Figure 1, the number of published datasets covering mast cells is substantially smaller compared to other hematopoietic cell lineages, illustrating the significant data scarcity. Deep learning methods including convolutional neural networks, encoder-decoder architectures, and recurrent frameworks have been proposed for the detection and study of MC in the context of mastocytosis, and the spread and development of MC can in principle be studied through microscopic image analysis using AI frameworks [12]. However, validated pipelines specifically targeting morphological classification of normal versus atypical MC in bone marrow aspirates, with direct alignment to WHO diagnostic thresholds, do not yet exist as established clinical tools.

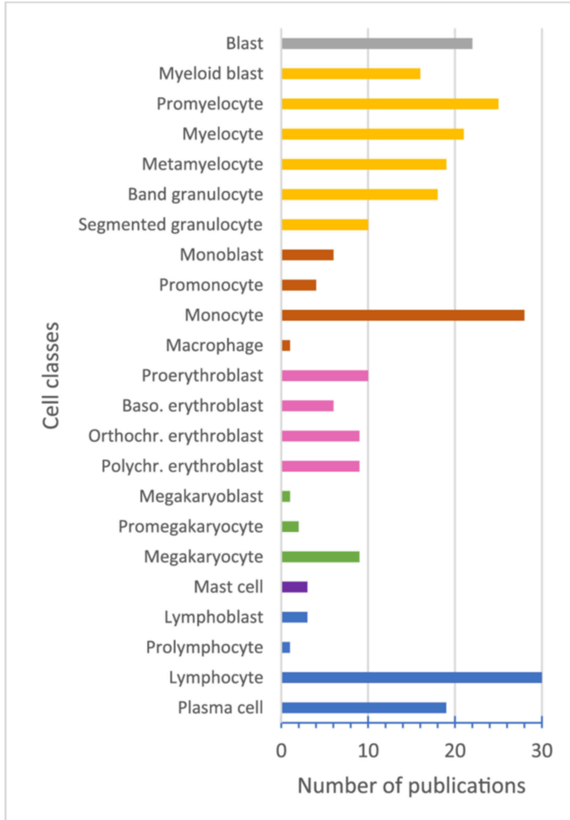

Figure 1: Publication distribution across cell lineages [9]

This gap is clinically significant. The minor diagnostic criteria for SM include the condition that at least a quarter of the MC in the bone marrow exhibit unusual characteristics, such as spindle-shaped or otherwise abnormal morphology [1]. Meeting or failing to meet this threshold has direct consequences for patient classification and treatment decisions. Manual cell counting is variable between observers and across institutions, and given the low absolute number of MC per sample, even small counting errors can shift a patient across a diagnostic boundary. An automated, reproducible, and transparent counting system therefore addresses a concrete unmet clinical need.

Taken together, the existing literature confirms that YOLO-based architectures are well-suited for cell-level detection in hematological microscopy, with performance in the range of 77-94% mAP achievable on well-represented cell types. However, no published study has specifically addressed the binary morphological classification of MC, atypical versus normal, in alignment with WHO diagnostic thresholds for SM. This thesis targets precisely this gap. It builds upon lessons learned from prior object detection experiments in related image analysis tasks [13], and applies cross-validation methodology to ensure that reported performance metrics are generalizable rather than optimistically overfitted to a single data split. The clinical ambition of the project is to provide a deployable tool that can be integrated into an existing hospital software environment which distinguishes it from purely academic explorations and places model reliability, interface usability, and system integration at the center of the evaluation criteria.

2. Methodology

2.1. Biology of Mast Cells and Systemic Mastocytosis

2.1.1. Mast Cells: Origin and Function

Mast cells are tissue-resident immune cells that play a central role in the body's innate defense system. They arise in the bone marrow, where maturation is controlled by stem cell factor binding to the receptor c-KIT, as well as by cytokines such as interleukin −3, −4, −9, and −10, which promote both differentiation and proliferation. Once mature, they migrate into neighbouring tissues and take up residence particularly in areas exposed to the external environment such as the gastrointestinal tract, respiratory mucosa, skin, and connective tissue surrounding blood and lymphatic vessels [14].

Structurally, MC contain a nucleus surrounded by hundreds of granules, which serve as storage compartments for chemical mediators. These include histamine, which opens blood vessels and allows fluid and immune cells into tissues; proteases such as tryptase, chymase, and carboxypeptidase, which break down connective tissue to allow for better passage of immune cells; and cytokines and chemokines, which signal immune cells to proliferate, adhere to vessel walls, and move to sites of injury or infection [15].

Beyond their well-known role in allergic reactions, MC also contribute to tissue repair and angiogenesis. MC's synthesize, store, and release hundreds of mediators in response to allergic, environmental, neuroimmune, stress, and toxic triggers, with some of these mediators released independently of degranulation, including angiogenic, fibrotic, and neurotoxic molecules [16]. This functional spectrum makes them relevant not only in allergy and host defense, but across a wide spectrum of inflammatory and neoplastic conditions.

2.1.2. Development of Systemic Mastocytosis

Systemic mastocytosis arises when MC undergo uncontrolled clonal proliferation driven by a somatic gain-of-function mutation. Around 95% of SM cases originate from a point mutation at codon 816 of the KIT gene, where aspartic acid is replaced by valine, referred to as KIT D816V. This substitution results in ligand-independent, constitutive activation of the KIT tyrosine kinase receptor, leading to unregulated MC proliferation, amplified growth, and increased survival [17].

2.1.3. Disease Subtypes and Progression

SM is not a single, uniform entity but a spectrum of disease subtypes that differ in clinical behavior and prognosis. In order of increasing severity these are indolent SM, smoldering SM, aggressive SM, SM with an associated hematological neoplasm (SM-AHN), and mast cell leukemia (MCL). The two current classifications differ in how they treat bone marrow mastocytosis: the 2022 International Consensus Classification places it within indolent SM as a subvariant, whereas the WHO 5th edition recognises it as a distinct subtype in its own right [18].

The majority of adult patients present with indolent SM, the most common form, marked by mediator-related symptoms, frequent skin involvement, and a low marrow MC burden (usually below 5 to 10% of cellularity), with an overall survival comparable to the matched general population [18]. Smoldering SM occupies an intermediate position, with a higher marrow burden (above 30%) and a greater risk of progression to aggressive disease, but without the organ damage that defines advanced SM [18].

2.1.4. Morphology and the Diagnostic Threshold

The distinction that matters for diagnosis, and the one this work automates, is morphological. A normal mast cell (Figure 3) is round, carries a single round nucleus with an inconspicuous nucleolus, and is filled with metachromatic granules so densely packed that they often obscure the nucleus. Neoplastic mast cells depart from this baseline: atypical cells (Figure 2) are elongated and spindle-shaped with hypogranulated cytoplasm, while the most immature forms, promastocytes (Figure 4 (B)) and metachromatic blasts (Figure 4 (C)), signal higher-grade disease [18]. This morphology is not merely descriptive but quantitatively diagnostic: at least 25% of all mast cells showing atypical or spindle-shaped morphology in the bone marrow smear or section constitutes a minor SM criterion [1], and a burden of at least 20% atypical mast cells in the bone marrow aspirate, characteristically immature, defines mast cell leukemia [18]. Because crossing these thresholds reclassifies a patient, a reliable count of atypical versus normal mast cells is the clinical signal itself, and producing that count automatically is the task the detector in this thesis is built to perform.

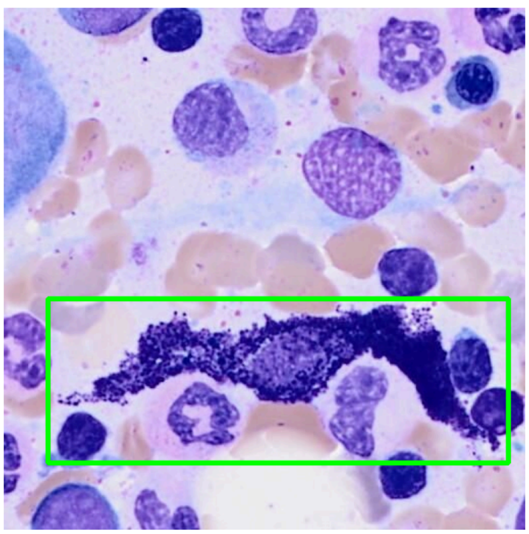

Figure 2: Sample of spindle-shaped cell from the 53-patient dataset

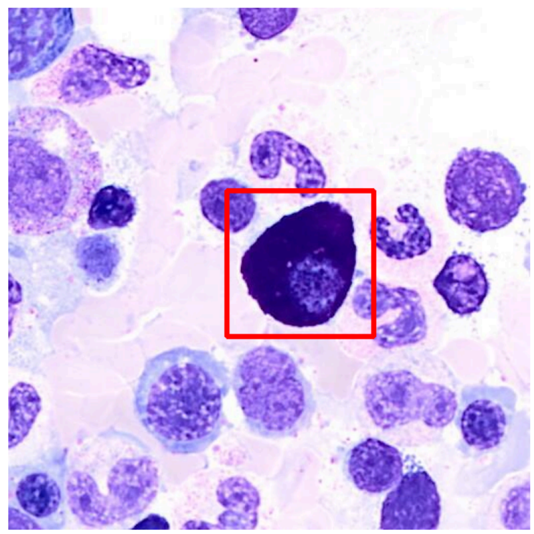

Figure 3: Sampe of normal-shaped cell from the 53-patient dataset

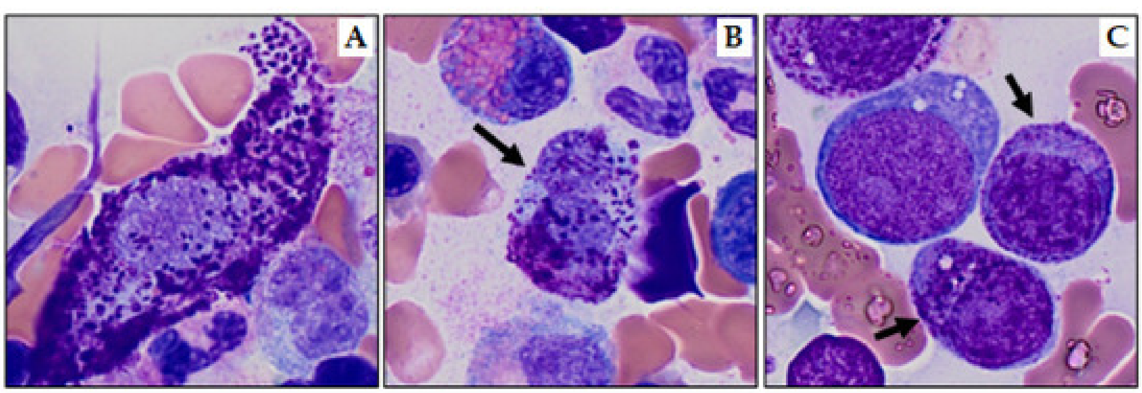

Figure 4: Atypical mast cells. (A) Spindle shaped (type I) mast cell, (B) bilobed promastocyte (type II; black arrow), and (C) metachromatic blasts (black arrows) [18]

2.1.5. SM as a Precursor to Other Hematological Neoplasms

Secondary genetic lesions on top of the KIT mutation, or in cooperation with a particular genetic background, are potentially required for the malignant transformation of indolent SM into more severe disease forms [19]. One such form is SM-AHN, which denotes systemic mastocytosis co-occurring with a separate associated hematological neoplasm; that associated neoplasm is myeloid in roughly 90% of cases, spanning myeloproliferative neoplasms (MPN), myelodysplastic syndromes (MDS), and myelodysplastic/myeloproliferative neoplasms such as chronic myelomonocytic leukemia (CMML) [20]. This myeloid neoplasm frequently carries the same KIT D816V mutation as the mast cell compartment, indicating a common clonal origin in a multipotent hematopoietic precursor [19].

The occurrence of the KIT mutation in bone marrow cell compartments other than MC is associated with progression to more aggressive forms of the disease and poor outcome in indolent SM. The extent of this multilineage involvement is therefore an important prognostic indicator, and its detection through the immunophenotypic profile of bone marrow mast cells may serve as a surrogate marker for prognostic stratification [21].

2.2. Technical

2.2.1. General Data Information

The dataset for this study consists of digitized bone marrow aspirate smear images provided by the USZ. All samples were collected from patients with confirmed or suspected systemic mastocytosis. Image tiles were extracted from whole-slide scans at a fixed resolution of 512x512 pixels, each displaying a sub-section of the whole smear. Two morphological classes were labeled for this study: atypical mast cells and normal mast cells, corresponding directly to the cell types relevant to the WHO minor diagnostic criterion for SM.

The corpus grew across four acquisition batches spanning 53 patients. The initial dataset covered a single patient (P1) and served as the pilot for the annotation and training pipeline. A second batch extended the corpus to 15 patients (P1-P15), contributing 1′564 annotated images with 1′288 atypical and 391 normal instances, reflecting a raw class ratio of approximately 3.3:1. A third batch added patients P16 through P53, bringing the full corpus to 53 patients (1′869 annotated images, 1′476 atypical, 511 normal, raw ratio 2.9:1). Because the training data derives almost entirely from confirmed SM patients, atypical is the dominant class throughout. At the P1-P8 scale the raw ratio was 14.4:1; by P1-P15 this had dropped to 3.3:1 due to the inclusion of patients P9 and P15 which are rich in normal annotations.

2.2.2. Data Acquisition and Annotation

The ground truth labeling of atypical and normal mast cells was carried out by Robin Kosche at USZ using CVAT (Computer Vision Annotation Tool [22]). Rather than annotating all 53 patients manually from scratch, an iterative model-assisted annotation strategy was used to make the labeling process tractable at the scale of tens of thousands of image tiles.

The first patient (P1) was annotated entirely by hand. Bounding boxes were drawn around each visible mast cell. This initial patient had only atypical cells. This dataset was used to train the initial domain-specific detection model, Model_{P1}, a YOLO11n checkpoint trained on P1 data.

For patients P2 through P15, CVAT's model inference integration was used with Model_{P1} as the backend. The model generated candidate bounding boxes for each image tile, which the annotator then reviewed, accepted, corrected, or rejected. Starting from this batch, normal cells were also annotated. Tiles where the model produced detections on regions that contained no mast cell were recorded separately in per-patient Excel files. These confirmed false positives were later incorporated into training as a curated set of hard-negative examples.

For patients P16 through P53, the same workflow was repeated using a second-generation model ($\text{Model}_{\text{P1}-\text{P15}}$), trained on the P1-P15 corpus, as the inference backend. It is worth noting that a subset of patients in this batch contained no annotatable mast cells at all: the model produced candidate detections, but all were rejected by the annotator as false positives. These patients contributed no positive labels to the corpus but supplied a large number of confirmed hard negatives. The annotations from the full P1-P53 corpus were then used to train a third-generation model, $\text{Model}_{\text{P1}-\text{P53}}$, completing the bootstrapped annotation cycle at the current project scope.

This iterative approach, where each annotation round produces a labeled dataset used to train a better model for the next round, allowed the corpus to grow from one patient to 53 without requiring fully manual annotation of all 53 cases. The confirmed false positives collected at each round provided a particularly valuable training signal: these are the specific tiles that the model of that generation found difficult to reject, making them targeted hard negatives for the next training cycle.

To our knowledge, this bootstrapped annotation strategy is not documented for mast cell detection in the existing literature. The systematic review by Ghete et al. [9], covering DL-based bone marrow cell classification from 2019 to 2024, found that the majority of published studies applied their models to pre-existing datasets rather than developing an annotation strategy to bootstrap the model. In this project, annotation and model development were tightly coupled by design: the clinical partner's annotation capacity was directly extended by each generation of model, and the confirmed false positives produced during annotation became training signal for the next generation. This coupling is what made a 53-patient corpus feasible within the scope of a single thesis.

The final corpus assembled across all three annotation rounds is summarized in Table 1.

Table 1: Full corpus summary across P1-P53.

Category	Count
Total annotated images	1′869
Atypical cell labels	1′476
Normal cell labels	511
Raw class ratio (Atyp:Norm)	2.9 : 1
Confirmed FP/Background images	4′940

2.2.3. Data Preprocessing

Two distinct preprocessing approaches were applied across the experimental timeline, with a significant revision midway through the project.

Pre-applied augmentation. For the initial experiments, a preprocessing pipeline was implemented using the albumentations library [23]. Each annotated image was transformed once before training with random flips, brightness and contrast adjustments, and affine rotations of up to 15 degrees. The augmented copies were saved to disk and used as fixed training input. This approach proved ineffective: because transforms were applied once and stored, every training epoch saw the same augmented copies. The augmented images extended the dataset by a fixed factor but provided no additional regularization benefit after the first epoch, which is the only epoch where the transforms are novel to the model. This limitation was identified during Phase 2 and motivated the switch to on-the-fly augmentation.

On-the-fly augmentation. Starting from Phase 3, augmentation was moved into the YOLO training loop by enabling augment=True in the model.train() call. This applies a fresh random transform to each image at every epoch, meaning oversampled copies in the training file list present different pixel content to the model across epochs. Two augmentation parameters required explicit configuration for this domain. The mosaic augmentation was disabled (mosaic=0.0): YOLO's mosaic mode composites four images into a single training sample, which is appropriate for multi-object scene datasets like "Common Objects in Context" (COCO) but is counterproductive for 512x512 bone marrow tiles, where the spatial relationship between a candidate cell and its immediate local context is part of the morphological signal the model needs to learn. Vertical flip probability was set to 0.5 (the YOLO default is 0), consistent with the biological reality that bone marrow mast cells have no canonical vertical orientation and that the model should not learn a vertical bias from the training data.

The rotation augmentation parameter (degrees) was treated as a tunable hyperparameter during Phase 3. A systematic tinkering matrix identified degrees=5 as optimal at the Phase 3 corpus scale, providing approximately 4-5 percentage points of improvement in normal-class recall over degrees=0. The effect is biologically justified: mast cells present in any orientation in aspirate smears, and moderate rotation augmentation regularizes the model without distorting morphological cues that distinguish atypical from normal.

2.2.4. Model Selection and Architecture

YOLO11 (Ultralytics) was selected as the object detection framework. The architecture performs detection and classification in a single forward pass, making it computationally efficient for integration into clinical workflows where inference throughput matters. The Ultralytics Python API provides a consistent interface for training, fine-tuning, evaluation, and inference, and supports checkpoint initialization from custom pretrained weights, which was central to the transfer learning strategy described below. Comparable studies applying YOLO-family models to cell detection in hematological microscopy report mAP50 values in the range of 77-94% on well-represented cell types [5] [10], placing the architecture in the performance range expected for this domain.

Among the available YOLO11 model sizes, the nano variant (yolo11n, 2.6M parameters) was selected. The selection was validated experimentally: a test using the large variant (yolo11l, 25.3M parameters [24]) on the P1-P8 corpus under identical training conditions produced R_{normal} = 0.695 compared to R_{normal} = 0.840 for yolo11n. The larger model's classification head, which is randomly re-initialized when training with a different number of output classes than the pretrained checkpoint, has proportionally more free parameters to adapt from scratch. With only 170-220 training images per Leave-One-Patient-Out (LOPO) fold, these parameters tend to overfit to the majority class rather than learn generalizable features for the rare normal class. The corpus size at this stage was insufficient for a 25.3M-parameter model; yolo11n was confirmed as the appropriate architecture for the available data.

Throughout the project, all training runs started from the generic COCO-pretrained yolo11n.pt checkpoint. The single exception was a set of experiments during Phase 2 in which $\mathrm{Model}_{\mathrm{P1}}$, a YOLO11n model previously trained on the P1 single-patient dataset (nc=1, one class), was used as the starting point for continued training on the two-class P2 dataset. The reasoning was that domain-specific pretraining might give the model a better starting point for bone marrow aspirate images compared to COCO features. In practice, this approach introduced a structural complication: because $\mathrm{Model}_{\mathrm{P1}}$ was trained with a single output class and the P2 training required two classes (atypical and normal), the Detect head's classification branch (cv3, the final 1x1 convolution layer) had to be discarded and randomly re-initialized. The bounding-box regression branch (cv2) transferred cleanly, but the randomly initialized cv3 layer produced noisy classification gradients in early training epochs. At the default learning rate of lr0=0.01, these gradients were large enough to partially overwrite the pretrained neck features, causing systematic underperformance rather than the expected improvement. The full account of this failure and its resolution is described in Section 2.2.5 "Model Training and Finetuning Strategy". After Phase 2, the project returned to yolo11n.pt as the standard base for all subsequent training.

2.2.5. Model Training and Finetuning Strategy

The training strategy developed across five phases, each responding to specific failures and insights from the previous phase. The following account describes the major decision points rather than every individual run; a complete run table is provided in (Section 3) "Results". The phase structure corresponds to successive expansions of the training corpus and methodology: Phase 1 covered a single pilot patient, Phase 2 moved to a two-class annotated dataset and explored data composition, Phase 3 introduced multi-patient LOPO cross-validation and on-the-fly augmentation on an 8-patient corpus, Phase 4 expanded to 15 patients, and Phase 5 targets the full 53-patient corpus. Individual experiment runs within each phase are identified by a label combining the phase number and a letter (e.g., P2-C, P3-A), which matches the experiment log maintained throughout the project.

Phase 1: Single-patient proof of concept and hyperparameter search.

Phase 1 covered a sequence of experiments on the P1 dataset (428 images, single patient, atypical class only) using 5-fold stratified cross-validation and the pre-applied albumentations augmentation pipeline described above. The dataset consisted of annotated positive tiles alongside hand-curated negative tiles (confirmed empty regions), all at 512x512 pixels.

The initial runs used yolo11n.pt (COCO pretrained). To test whether a larger model would learn better features for this domain, yolo11l was then substituted and run in three separate configurations, including extended epoch budgets (200 epochs) and added augmentation. In each case, the large model did not outperform yolo11n on this single-patient dataset, and the project returned to yolo11n for all further Phase 1 work. This result prefigured P3-E; the mechanism is described in Section 2.2.4 "Model Selection and Architecture".

With yolo11n established as the appropriate architecture, hyperparameter tuning was run using YOLO's built-in model.tune() function across three increasingly intensive rounds: 30 iterations x 50 epochs, 300 iterations x 50 epochs, and finally 300 iterations x 100 epochs. The output of the final tuning round was a best_hyperparameters.yaml file with optimized values including lr0=0.0062, momentum=0.791, mosaic=0.916, and degrees=0.0. A training run applying these tuned hyperparameters completed Phase 1.

The best_hyperparameters.yaml produced by the Phase 1 tuner is significant beyond Phase 1: it was later applied to fine-tuning on P2 data (P2-E) and caused catastrophic forgetting, collapsing performance across all five folds. The failure was later attributed to a recipe mismatch: the tuner had optimised on a from-scratch yolo11n training run, and its output (particularly the high mosaic value and degrees=0.0) was calibrated for scratch training from COCO, not for fine-tuning in this clinical domain. The Phase 1 yaml was permanently retired from the fine-tuning pipeline as a result.

Following the 5-fold training, an inference run was conducted on the full first-patient slide image set, approximately 20′000 tiles, excluding all training images by filename. The resulting confidence distribution provided an early look at the model's behavior at deployment scale and informed the direction of subsequent annotation and training work.

Phase 2: The domain-specific checkpoint experiment and the learning rate problem.

Phase 2 began with two runs using the standard yolo11n.pt base on the annotated P2 dataset (P2-A, P2-B), but both were corrupted by a concurrent SLURM job collision described below. The third run (P2-C) introduced Model_{P1} as the starting point, testing whether the P1-trained domain-specific model would give the detector a better foundation than COCO-pretrained weights. The result was a characteristic failure pattern: high precision combined with very low recall, the signature of a model that makes few predictions and misses most cells. The cause was the interaction between the randomly re-initialized cv3 head and the default learning rate of lr0=0.01: the noisy gradients from the freshly initialized classification branch were large enough to partially overwrite the pretrained neck features during early epochs, leaving the model worse than yolo11n.pt would have been from scratch.

Reducing the learning rate to lr0=0.001 recovered the performance lost to this overwriting effect. Freezing the first ten backbone layers (freeze=10) was introduced alongside the learning rate change to prevent the small dataset from further disrupting features that COCO pretraining had established. This combination (lr0=0.001, freeze=10) defined the reference configuration for the remainder of Phase 2. However, the conclusion drawn from the full Phase 2 experiment set was that using Model_{P1} as a starting point added complexity without a measurable benefit over yolo11n.pt, and all subsequent phases returned to the standard COCO-pretrained base.

Before expanding to the full P1-P15 corpus that would produce the second model version, a 3x4 factorial design-of-experiment (DoE) grid was run to identify the best data composition strategy at Phase 2 scale. The goal was practical: the hyperparameter and data configuration choices made here would carry forward into the P1-P15 training, and a better second model version meant fewer manual corrections for the annotation partner at USZ when working through patients P16-P53. The grid varied confirmed FP negative oversampling (1x, 2x, 3x) and random background tile ratio (0, 1, 2, 3) across twelve configurations. The mAP50 spread across all twelve DoE configurations was smaller than the cross-fold standard deviation within any single run. Data composition manipulations at single-patient scale could not produce statistically interpretable performance differences; the limiting factor was data scarcity rather than class balance. Background tiles uniformly hurt performance at every FP oversampling level, and this finding was consistent enough to permanently set the random background ratio to zero in all subsequent runs. The DoE finding reframed the problem: improving recall required more patients and better augmentation, not a different ratio of existing data categories. The transferable lesson for rare-event medical imaging is discussed in Section 4 “Discussion”.

Phase 2 also exposed an infrastructure failure specific to the High-Performance-Computing (HPC) environment. Submitting two SLURM jobs concurrently caused both jobs to share the same temporary workspace directories, because directory paths were derived from fold indices without job-level namespacing. Concurrent writes to shared directories corrupted three of five folds in each job, detectable by fold results that were bit-for-bit identical between the two affected runs. The fix applied $SLURM_JOB_ID namespacing to all temporary directories and derived the output project directory name from the job name set at sbatch submission time, making concurrent jobs structurally independent.

Phase 3: Multi-patient LOPO and on-the-fly augmentation.

Phase 3 restructured the training and evaluation design around three simultaneous changes: expansion to 8 patients, adoption of LOPO cross-validation, and the switch to on-the-fly augmentation.

LOPO was adopted to eliminate the patient-level leakage present in stratified k-fold splits: holding out all images from one patient per fold makes each fold an evaluation of genuine cross-patient generalization. The full rationale and its relation to external validation standards are in Section 2.2.6 “Model Testing”.

The combined effect of these three changes was substantial. The Phase 3 baseline run (P3-A: 8 patients, yolo11n.pt, augment=True, mosaic=0.0, cls=1.0) achieved the strongest results seen to that point in the project, with cross-fold standard deviation markedly lower than the Phase 2 DoE despite the harder evaluation protocol: the more diverse training corpus more than compensated for the stricter holdout design.

With the same goal of finding the best possible training recipe before the P1-P15 training that would produce the second model version, a tinkering matrix of 14 parameter combinations explored rotation degrees (0, 5, 10), distributed focal loss weight (dfl) (1.5, 2.0), backbone freeze depth (0, 10), and label smoothing (0.0, 0.1). The production recipe confirmed here would be directly applied to the P1-P15 training: a higher-quality second model version meant the annotation partner at USZ would need fewer manual corrections when processing patients P16-P53. Rotation degrees was the dominant factor: degrees=5 consistently improved R_{normal} over degrees=0 across all freeze and dfl combinations. Mast cells present in all orientations in aspirate smears, and moderate rotation augmentation directly targets this. The dfl parameter showed a consistent directional effect: dfl=2.0 regressed R_{normal} relative to dfl=1.5 across all other parameter combinations, suggesting that increasing localisation loss weight tilts model capacity away from the classification discrimination needed for the rare normal class. Freeze depth confirmed the Phase 2 finding: freeze=10 outperformed freeze=0 in every recall-relevant comparison. Label smoothing proved inert in this Ultralytics version, likely because the implementation does not propagate the parameter through the classification loss path.

The production recipe confirmed by the tinkering matrix was: degrees=5, dfl=1.5, freeze=10, lr0=0.001, cls=1.0, augment=True, mosaic=0.0, flipud=0.5.

An automated hyperparameter search using YOLO’s built-in model.tune() was run on the production recipe (30 iterations, 50 epochs, patient P4 as holdout, mAP50-95 as fitness). The tuner converged on degrees=0.0 and dfl=2.47, both directions that the tinkering matrix had identified as harmful to R_{normal}. The explanation is structural: a single-fold tuner optimizing mAP50-95 on a P4 holdout (an atypical-dominant patient) maximizes atypical-class performance at the expense of normal-class generalization. Single-fold automated tuning is blind to cross-patient per-class recall balance. The production recipe from the tinkering matrix was retained.

A model-size comparison (P3-E) substituted yolo11l for yolo11n under identical conditions on the P1–P8 corpus and produced R_{normal} = 0.695, the lowest value recorded across the entire project (mechanism in Section 2.2.4 “Model Selection and Architecture”).

The production recipe confirmed by Phase 3 was carried forward and applied to train on the full P1-P15 corpus, producing the second model version ($\mathrm{Model}_{\mathrm{P1-P15}}$) that the annotation partner at USZ used for patients P16-P53.

Phase 4: Corpus expansion to 15 patients.

Phase 4 expanded the training corpus to 15 patients (1′564 images, 1′288 atypical, 391 normal, 2′464 background images including FP; raw ratio 3.3:1, effective ratio 1.5:1 after automatic per-patient oversampling). The most significant addition was patient P9, which contributed 195 normal annotations, more than doubling the total normal count relative to the P1-P8 corpus. A 15-fold LOPO protocol was applied.

The first Phase 4 run (P4-A) was configured with degrees=0 due to an omitted configuration variable; the second run (P4-B) applied the full production recipe including degrees=5. The delta between the two runs was within the cross-fold standard deviation on all metrics, indicating that the benefit of degrees=5 observed at P1-P8 scale either diminishes with a more diverse 15-patient corpus or is within measurement noise at this scale. A freeze ablation run (P4-C, freeze=0) produced results essentially equivalent to P4-B across all metrics, confirming that backbone freezing remains appropriate and the P3-B recipe transfers to the larger corpus without re-tuning. Full per-fold results for P4-A, P4-B, and P4-C are reported in Section 3 “Results”.

Phase 5: Full corpus (P1-P53).

The Phase 5 training run (P5-A) applied the confirmed production recipe (degrees=5, dfl=1.5, freeze=10, lr0=0.001, cls=1.0) with fully automatic per-patient oversampling targeting a 2:1 atypical:normal ratio across a LOPO design over all 35 annotated patients.

2.2.6. Model Testing

Performance estimates in this thesis are reported under two different evaluation protocols depending on the phase. Phase 1 and Phase 2 used stratified k-fold cross-validation (5 folds), where images were split randomly across folds with class stratification. From Phase 3 onward, the evaluation protocol was changed to LOPO, where all images from one patient are withheld from training entirely and used exclusively for evaluation. The switch was motivated by a systematic bias identified in the Phase 1 and Phase 2 results: stratified k-fold splits on small medical imaging datasets allow patient-level texture and staining style to leak from training into validation, artificially improving validation metrics without reflecting true cross-patient generalization. LOPO results and stratified k-fold results are therefore not directly comparable.

Choosing LOPO over stratified k-fold directly addresses a gap identified in the broader literature. Ghete et al. [9] found that performance metrics across published bone marrow cell classification studies are not directly comparable in part because most studies evaluate on subsets of their own data, with few reporting results on patient-held-out splits and fewer still validating externally. LOPO is the strongest within-dataset approximation of external validation available without access to a second institution’s data: each fold evaluates the model on a patient it has never seen in any form, under the same conditions a deployed system would face in routine clinical use. The cross-fold standard deviation reported in this thesis therefore reflects genuine inter-patient biological variability rather than statistical noise from random sampling, making the performance estimates more conservative and more informative than single-split or random k-fold results typically reported in the field.

Each fold produced a training split, a validation split used only for early stopping during training, and a test split consisting of the held-out patient’s images. The metrics reported throughout this thesis refer to the test split, if not mentioned otherwise. Across folds, the mean and standard deviation of each metric were computed. The cross-fold standard deviation reflects genuine inter-patient biological variability; patients with very small annotated sets (for example, P13 with 7 images) produce test metrics with extreme sensitivity, where a single missed cell changes recall by more than 14 percentage points.

A per-class recall extraction issue affected single-class holdout folds through Phase 4; affected folds are excluded from per-class mean calculations throughout. The issue is fully described, with the fix applied in Phase 5, in Section 4.1.2 “Measurement Notes”.

2.2.7. Evaluation Metrics

Model performance was assessed using standard metrics from object detection. Each detection outcome is first classified into one of four categories based on whether the model’s predicted bounding box overlaps sufficiently with a ground-truth annotation (at a minimum Intersection over Union of 0.50):

- **True Positive (TP):** the model correctly detected a mast cell and the predicted box sufficiently overlapped the ground-truth annotation.
- **True Negative (TN):** the model correctly produced no detection in a tile containing no annotated mast cell.
- **False Positive (FP):** the model predicted a mast cell where the ground truth showed none (a false alarm).
- **False Negative (FN):** the model failed to detect a mast cell that was present in the ground truth (a missed cell).

Because the confusion-matrix is built per class, a detection that is correctly localized but assigned the wrong class is not a single error but two: it is placed in an off-diagonal cell, lowering the recall of its true class and the precision of the predicted class at once, so it behaves as a FN for the correct class and a FP for the wrong one [25].

From these, the following performance indicators were computed [26]:

- **Precision** measures the proportion of predicted detections that were correct:

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall** measures the proportion of ground-truth mast cells that the model successfully detected:

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1-Score:** The F1-Score represents the harmonic mean of Precision and Recall, providing a single metric that balances both factors.

$$F1 - Score = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$$

- **mAP50** (Mean Average Precision at IoU 0.50) is the mean of the per-class Average Precision computed at a bounding-box overlap threshold of 50%. AP is the area under the precision-recall curve; mAP50 averages this across all classes:

$$\text{mAP50} = \frac{1}{N}\sum_{i=1}^{N} AP_{i@50}$$

- **mAP50-95** extends mAP50 by computing AP (Average Precision) at ten evenly spaced IoU thresholds from 0.50 to 0.95 (`torch.linspace(0.5, 0.95, 10)`) and averaging the result across thresholds and classes [25]. It penalises imprecise bounding box localisation more heavily than mAP50.

mAP50 is used as the standard benchmark for run-to-run comparison across the experimental timeline, providing a single-number summary that captures both detection and classification quality. mAP50-95 is reported where available but assigned less interpretive weight; localisation precision at high IoU thresholds is clinically less relevant than detection recall for this task. Precision is monitored throughout but treated as a secondary metric.

Two per-class recall metrics are tracked separately throughout the experiments: R_{atypical} (recall for the atypical class) and R_{normal} (recall for the normal class). Both are clinically necessary because the WHO criterion depends on the ratio of atypical to total detected mast cells. Undercounting normal cells inflates this ratio, potentially triggering a false-positive SM classification in borderline patients. Tracking both classes independently, rather than reporting a single aggregate recall, is a direct consequence of aligning the evaluation to the diagnostic question rather than to standard object detection benchmarking practice. Published studies in adjacent domains, such as general white blood cell subtype classification, typically optimize for aggregate mAP or F1 across all classes [10]; this thesis treats R_{normal} as a primary metric in its own right because the WHO threshold makes the normal count as diagnostically load-bearing as the atypical count. R_{normal} received particular experimental attention because normal is the minority class in the training corpus, consistently exhibiting lower recall and higher cross-fold variability than R_{atypical}.

Recall is the primary metric for this project. A missed mast cell (false negative) causes the atypical fraction to be underestimated, potentially placing a patient below the WHO ≥25% diagnostic threshold and delaying diagnosis of SM. A false positive is less consequential, though not negligible: the clinical workflow includes a manual review step where false alarms can be dismissed, and the diagnostic impact of a false positive depends on how far the true ratio is from the ≥25% threshold. In a patient with a strongly atypical cell population, a moderate false positive rate does not change the clinical conclusion because the atypical fraction remains clearly above the threshold. False positives become clinically relevant primarily in borderline cases where the ratio is close to ≥25%, and it is precisely in those cases that the manual review step is most likely to be exercised. Because the application computes the displayed ratio before that review, the false-positive rate is still controlled at deployment through the operating threshold rather than treated as free; the mechanism and its quantification are analysed in Section 4.3 “Error Analysis”.

Two confidence-dependent conventions apply throughout the reported results. The mAP metrics integrate over the full confidence range and are threshold-independent by construction. Precision, recall, and the per-class recalls (R_{atypical}, R_{normal}) are point estimates and are reported at the confidence value that maximises the mean F1 score across both classes, which is the operating point Ultralytics’ val() selects internally [25]. These point estimates are therefore not the maxima of their respective curves at near-zero confidence, but the values at the F1-optimal threshold; the deployment operating point ultimately used at USZ is set separately from the P11 calibration cohort (Figure 22).

2.2.8. UI Creation

Gradio Prototype

An initial web-based interface was developed using Gradio [27] to support early-stage result visualization and model evaluation during development. The prototype provided two functional tabs. The Analysis tab ran batch inference on image directories and displayed tiles with YOLO detection overlays (Figure 5), producing labeled output images. The Research tab supported ground-truth versus prediction comparison (Figure 6, Figure 7), applying a lower confidence threshold to maximize recall sensitivity during evaluation. A statistics dashboard (Figure 8) aggregated per-class detection counts and displayed the atypical fraction relative to the WHO ≥25% threshold. The Gradio prototype served its purpose as a rapid development tool and was later superseded by the desktop application.

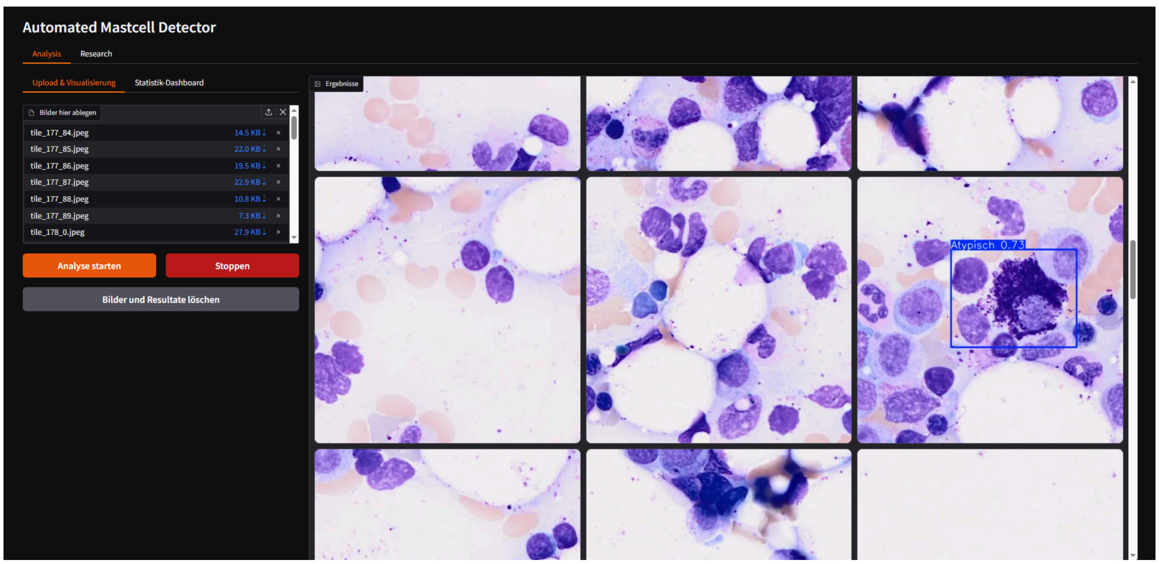

Figure 5: Analysis tab of the Gradio prototype showing uploaded image tiles with YOLO detection overlays.

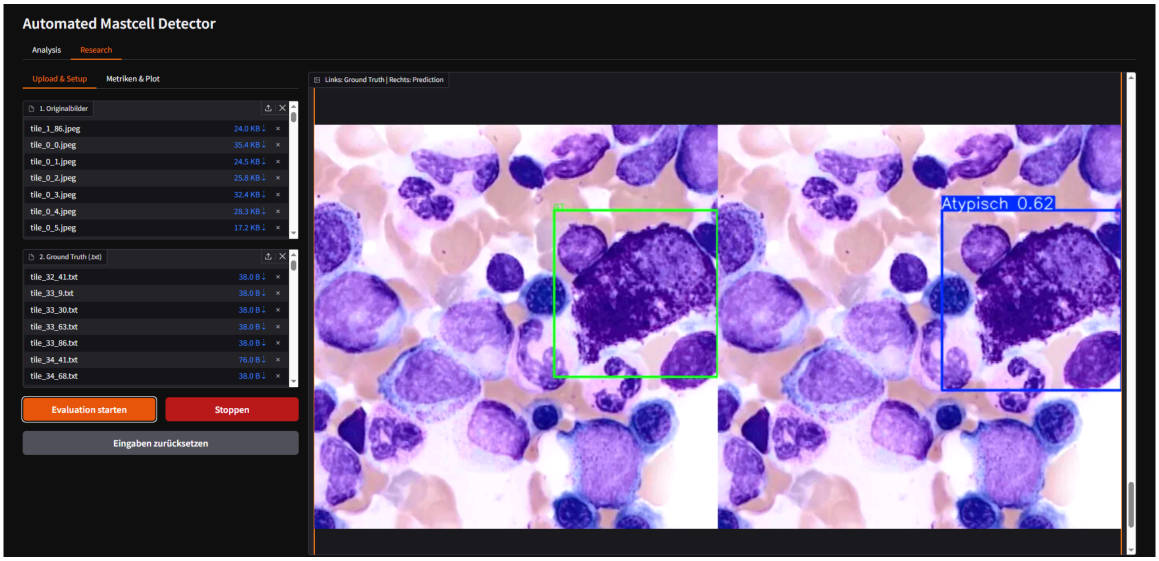

Figure 6: Research tab showing ground-truth annotations alongside model predictions for direct visual comparison.

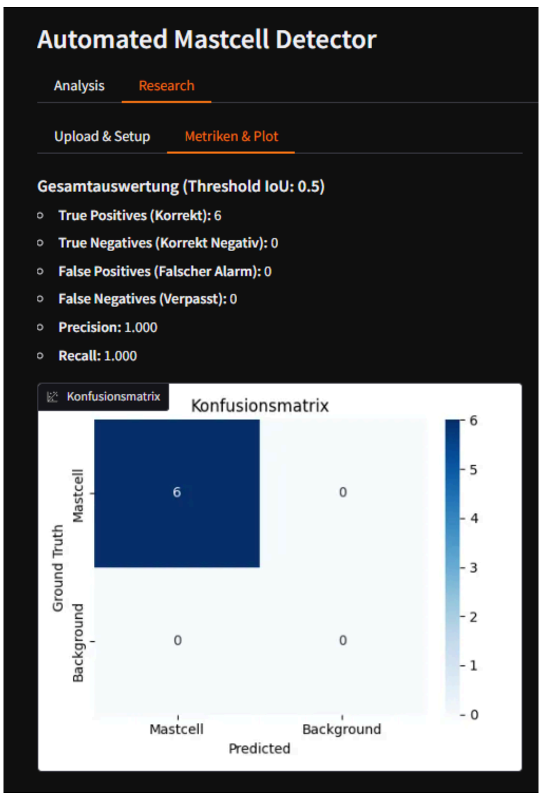

Figure 7: Research tab metrics output: precision-recall curves and performance plots generated from the ground-truth evaluation run.

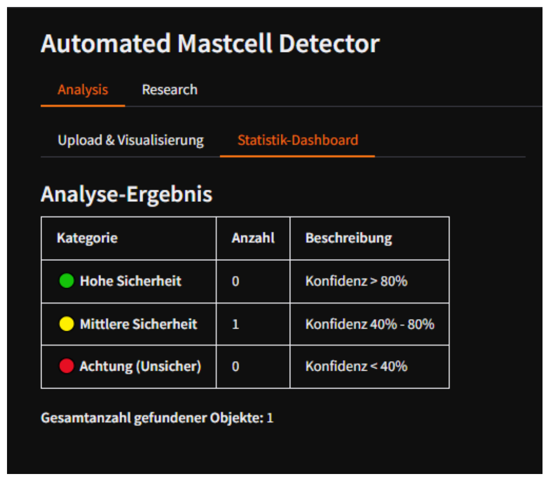

Figure 8: Statistics dashboard of the Gradio prototype showing per-class detection counts and the atypical fraction relative to the WHO ≥25% diagnostic threshold.

Desktop Application

As the requirements for clinical deployment became clearer, it became apparent that a browser-based interface was not adequate for the actual usage context at USZ. A full patient slide scan produces approximately 20′000 image tiles, and a Gradio-based interface was not designed for streaming inference at that scale. Cinderella, the cell classification software already in use at USZ, was considered as an integration target, but embedding the pipeline into an existing hospital codebase within the thesis timeline was not feasible: familiarizing oneself with a live clinical system, together with the security validation and change management processes required for hospital software integration, extended well beyond the project scope.

A native desktop application (MastCellDetector [28]) was developed as the primary clinical deliverable. The application was built with PyQt and packaged using PyInstaller into a self-contained executable that requires no Python environment on the target machine. While the primary deployment target is a Windows workstation at USZ, PyInstaller equally supports Linux builds, making the same application distributable on Linux servers or HPC systems.

The intended workflow for a USZ analyst proceeds as follows. The analyst opens a local folder containing a patient's image tiles. The sidebar provides controls for confidence threshold, IoU non-maximum suppression threshold, and hardware selection (GPU with automatic VRAM-based batch size recommendation, or CPU fallback). Running inference streams YOLO-format label files to disk alongside each image in real time, so results are preserved if the run is interrupted. The gallery view (Figure 9) displays all images with filter tabs for "All images", "With detections" (Figure 10), "No detection", "Low confidence", "Atypical only", and "Normal only". This allows a haematologist to navigate directly to the cases most likely to need manual review rather than scrolling through tens of thousands of tiles.

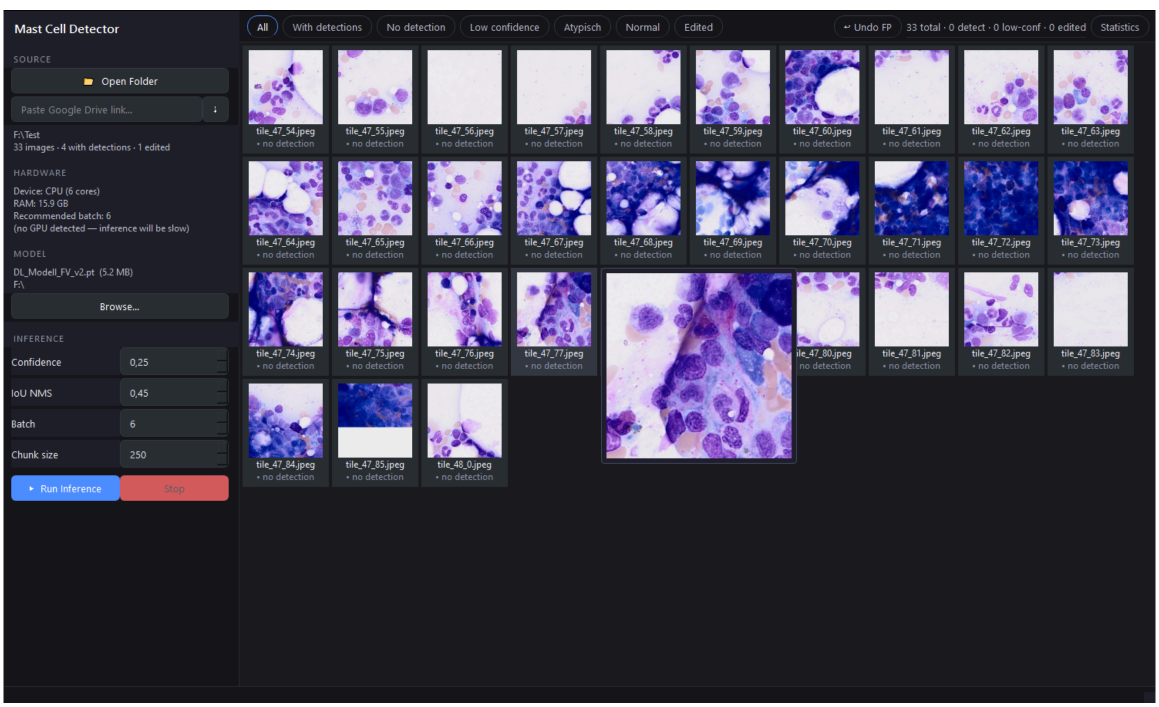

Figure 9: Gallery view of the desktop application showing all image tiles for a patient folder with sidebar controls for inference configuration.

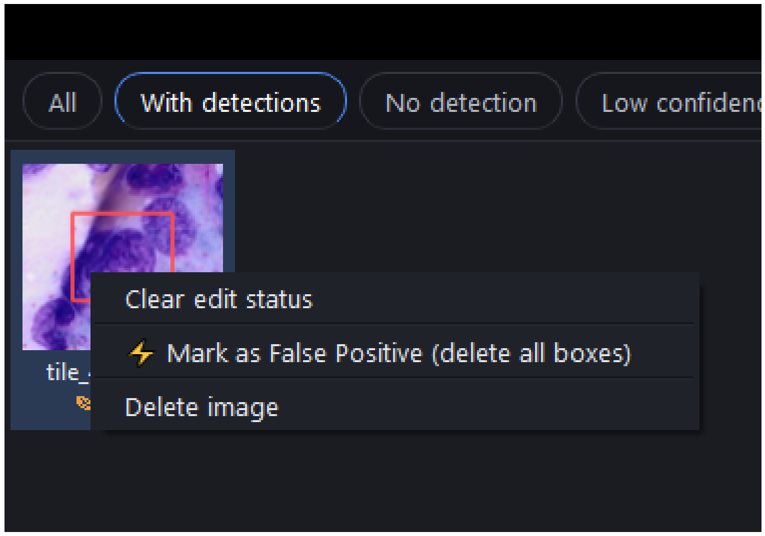

Figure 10: Gallery filtered to tiles with detections, allowing rapid navigation to images where the model found mast cells.

An annotation editor is embedded in the gallery. Clicking any thumbnail opens the image with interactive bounding boxes (Figure 9): boxes can be repositioned by dragging, resized by their corner handles, and removed with the Delete key. Hovering over a thumbnail magnifies it for closer inspection; on an empty tile the plain image is enlarged, while on a tile that carries a detection the magnification centers on the bounding box, so the candidate cell can be judged without opening the full editor. New boxes are drawn by pressing the "A-Key" on the keyboard and dragging out a region, with the class chosen from a dropdown. Changes are written to the corresponding label file with "Ctrl+S" or automatically on navigation to the next image.

The editor is built around the review workflow of a rare-event task, where background tiles vastly outnumber true cells and rejecting false positives is the most frequent action. Clearing the detections on a tile leaves an empty label file, which marks the tile as a confirmed negative for any future retraining cycle; the Delete key therefore doubles as a one-press reject that writes this confirmed-negative label in a single keystroke. Edited files are protected from being overwritten by subsequent inference runs, so curated ground truth is never lost when the model is re-run. The edited flag can also be cleared, in which case no label file is written for the tile and it reverts to its unreviewed state, again subject to overwrite on the next inference run (Figure 10).

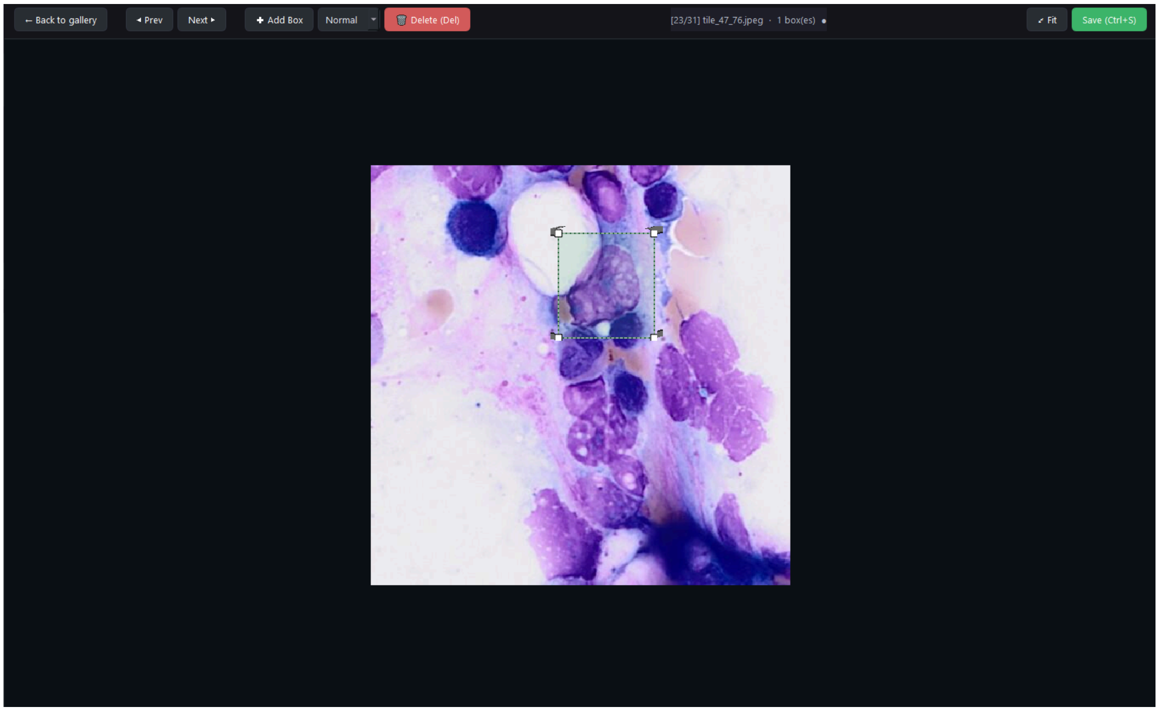

Figure 11: Annotation editor with interactive bounding boxes; boxes can be added, moved, resized, or deleted directly in the application without external tooling.

The Statistics tab (Figure 12) aggregates per-image detection counts into a patient-level summary that displays the total detected mast cells, the atypical fraction as a percentage, and a visual marker relative to the WHO ≥25% diagnostic threshold. This summary can be exported for clinical documentation or further analysis.

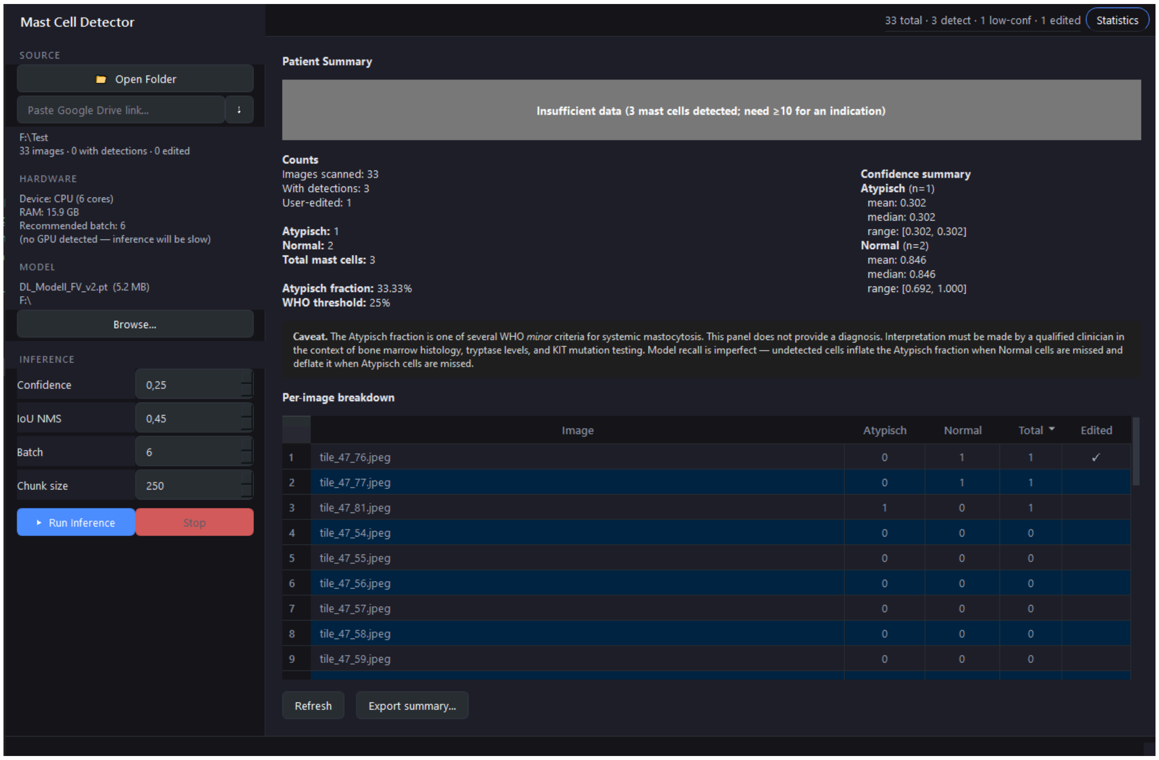

Figure 12: Statistics tab aggregating per-image detections into a patient-level summary with the atypical fraction displayed against the WHO ≥25% diagnostic threshold.

2.2.9. Implementation in USZ Infrastructure

The primary deliverable of this project is the MastCellDetector desktop application, distributed as a self-contained Windows application. The application operates entirely offline after installation, which satisfies a practical requirement for deployment in hospital network environments where internet access from clinical workstations is restricted or prohibited.

The final deployment model (P6-A) was trained on 1′832 images across 34 annotated patients (P1–P53 minus P11). Weights: `hpc/model_weights/DL_Modell_FV_v3.pt`. This is the model that will be shipped with the desktop application.

The annotation editing and label export functionality in the application also addresses a secondary deliverable: the confirmed false positives and corrected labels produced during clinical use feed directly back into the training pipeline as the next generation of hard-negative examples, continuing the iterative annotation strategy established from Phase 1 onward.

The Linux build capability opens a second deployment path beyond the clinical workstation. A Linux build of MastCellDetector can be deployed on a server or HPC node at USZ, allowing the tool to be integrated directly into the institution's scan-to-inference-to-evaluation pipeline: whole-slide images are scanned, tiles are passed through the model at server-side GPU throughput, and the resulting label files and statistics export are produced without requiring a Windows workstation in the loop. This makes the same application suitable for both interactive clinical review and automated batch processing, depending on the deployment context. The source code of the application is publicly available [28].

3. Results

The following training hyperparameters are referenced throughout this section:

- **`lr0`** — initial learning rate. Controls the step size at the start of fine-tuning; too high a value causes catastrophic forgetting of pretrained weights.
- **`freeze`** — number of backbone layers frozen during training. `freeze=10` locks the YOLO11n backbone and trains only the detection head; `freeze=0` allows full fine-tuning.
- **`degrees`** — maximum rotation angle for on-the-fly augmentation. Increases orientation diversity in the training distribution.
- **`dfl`** — Distribution Focal Loss weight. Scales the bounding-box regression loss relative to the classification loss; higher values shift the model toward localisation accuracy.
- **`cls`** — classification loss weight. Scales the class-prediction loss; used here to compensate for class imbalance between atypical and normal.
- **`mosaic`** — probability of mosaic augmentation (four tiles composited into one image). Disabled (`mosaic=0.0`) throughout because mosaic scrambles single-cell 512 px crops.
- **`flipud`** / **`fliplr`** — vertical and horizontal flip augmentation probabilities.
- **`FP_NEG_OVERSAMPLE`** — repeat factor for confirmed false-positive tiles in the training split. Values above 1 up-weight hard negatives.
- **`BG_RATIO`** — ratio of unannotated background tiles added as negatives per training image.

3.1. Model Results and Statistical Evaluation

Performance metrics for Phase 3 and Phase 4 are reported from the end-of-run evaluation of each LOPO fold, computed on the held-out patient that served as that fold's test set. Phase 1 and Phase 2 results derive from the held-out test partition of each fold under 5-fold stratified cross-validation and are not directly comparable to the Phase 3–4 LOPO estimates, for the reasons described in Section 2.2.6 "Model Testing": the stratified folds partition at the image level, whereas LOPO partitions at the patient level. All values quoted in this section refer to cross-fold means unless stated otherwise.

Phase 1

On the held-out test partitions, the P1-baseline reached mAP50-95 = 0.543 ± 0.046, precision = 0.770 ± 0.042, and recall = 0.810 ± 0.031 across the five stratified folds. The cross-fold spread was moderate: Fold 1 was strongest at mAP50-95 = 0.580 and Fold 2 weakest at 0.476, a range narrow enough that no single split dominates the mean. No per-class breakdown is available, as the first hand-annotated version of the Phase 1 corpus contained only atypical cells; the two normal annotations later present in P1 were added during model-assisted re-annotation and were not yet in the data at this stage.

Phase 2

The 3×4 factorial DoE grid (11 cells, FP oversampling ∈ {1×, 2×, 3×} crossed with background ratio ∈ {0, 1, 2, 3}) found no statistically distinguishable difference across compositions: cell-level mAP50-95 variation was smaller than the within-cell cross-fold standard deviation. Background tiles decreased performance at every FP oversampling level. The fp=1, bgr=0 configuration (P2-H) was the grid winner at every metric, achieving mAP50 = 0.604 ± 0.202, mAP50-95 = 0.494 ± 0.166, precision = 0.934 ± 0.057, and recall = 0.537 ± 0.210.

Phase 3

The structural pivot to 8-fold LOPO on 8 patients with on-the-fly per-epoch augmentation produced the largest single performance step in the project. P3-A achieved mAP50 = 0.844 ± 0.147, mAP50-95 = 0.718 ± 0.137, precision = 0.764 ± 0.138, and recall = 0.840 ± 0.122, a 30.3 percentage point improvement in recall over P2-H (0.537) as well as R_{atypical} = 0.884 ± 0.170 and R_{normal} = 0.795 ± 0.164. The cross-fold standard deviation dropped from 0.166 to 0.137 despite the harder evaluation protocol.

The tinkering matrix (P3-B, 14 of 24 cells completed) identified `degrees=5` rotation augmentation as the dominant training-time lever. The winning configuration (`degrees=5, dfl=1.5, freeze=10, label_smoothing=0.0`) achieved mAP50 = 0.864 ± 0.134, mAP50-95 = 0.741 ± 0.122, precision = 0.812 ± 0.153, recall = 0.866 ± 0.131, R_{atypical} = 0.892 ± 0.173, and R_{normal} = 0.840 ± 0.176. `dfl=2.0` consistently regressed R_{normal} relative to `dfl=1.5` across all degree and freeze combinations; `freeze=0` underperformed `freeze=10` in every recall-relevant slice; label smoothing was empirically inert (bit-identical fold metrics at `ls=0.0` and `ls=0.1`). These findings defined the production recipe (`degrees=5, dfl=1.5, freeze=10, lr0=0.001, cls=1.0`) adopted for all subsequent phases.

The automated hyperparameter tuner (P3-C, 30 iterations × 50 epochs, single P4 holdout, mAP50-95 fitness) converged on `degrees=0.0` and `dfl=2.47`. The production recipe from P3-B was retained.

The model-size comparison (P3-E, yolo11l) produced mAP50 = 0.855 ± 0.103, mAP50-95 = 0.745 ± 0.095, precision = 0.778 ± 0.134, recall = 0.785 ± 0.103, R_{atypical} = 0.875 ± 0.169 and R_{normal} = 0.695 ± 0.144, the lowest R_{normal} across all runs and 14.5 percentage points below P3-B.

Phase 4

Adding patients P9–P15 (449 new images, 313 new normal annotations) expanded the corpus to 1′552 images and 15 LOPO folds. P4-A (degrees=0, the production recipe with the rotation variable inadvertently unset) achieved mAP50 = 0.903 ± 0.109, mAP50-95 = 0.793 ± 0.106, precision = 0.851 ± 0.097, recall = 0.904 ± 0.120, R_{atypical} = 0.913 ± 0.110, and R_{normal} = 0.890 ± 0.194. The R_{normal} gain from corpus expansion (+9.5 pp over P3-A) exceeded the largest single hyperparameter gain observed in Phase 3 (+4.5 pp from `degrees=5` in P3-B).

P4-B (full production recipe with `degrees=5` explicitly set) produced mAP50 = 0.887 ± 0.120, mAP50-95 = 0.789 ± 0.118, precision = 0.826 ± 0.131, recall = 0.895 ± 0.132, R_{atypical} = 0.900 ± 0.174, and R_{normal} = 0.882 ± 0.193. All P4-A vs P4-B deltas were within 1.6 percentage points. The freeze ablation (P4-C, `freeze=0`) produced mAP50 = 0.883 ± 0.123, mAP50-95 = 0.782 ± 0.124, precision = 0.863 ± 0.087, recall = 0.893 ± 0.135, R_{atypical} = 0.896 ± 0.172, and R_{normal} = 0.882 ± 0.193, all within 0.4 percentage points of P4-B. P4-B is the primary reported result for the P1–P15 phase, as all training parameters are explicitly set.

Phase 5

Phase 5 evaluated the pipeline at full deployment scale across all 35 annotated patients from the 53-patient corpus. Two protocols were applied: a 35-fold LOPO run (P5-A) and a grouped 5-fold CV (P5-B). P5-B is the primary generalization result; P5-A is an optimistic upper bound documented for methodological completeness.

P5-A: 35-fold LOPO (optimistic upper bound)

The production recipe was applied across all 35 annotated patients. Of the 35 folds, 34 completed within the 48-hour SLURM allocation; Fold 17 (P37 holdout) was retrained independently and evaluated in a post-hoc inference pass (see Measurement Notes).

Across 35 folds, post-hoc evaluation reported mAP50 = 0.948 ± 0.146, mAP50-95 = 0.858 ± 0.120, precision = 0.925 ± 0.128, recall = 0.951 ± 0.084, R_{atypical} = 0.920 ± 0.129 (n=20 folds with atypical ground truth), and R_{normal} = 0.941 ± 0.119 (n=29 folds with normal ground truth). Of the 35 folds, 19 reported mAP50 at the metric ceiling (0.995), corresponding to patients from the P16–P53 acquisition batch who contributed only one to three annotated images each. On a one-image test set, a single correct detection produces recall = 1.000 regardless of model generalization capacity, inflating the aggregate mean.

P5-B: Grouped 5-fold CV (primary Phase 5 result)

The 35 annotated patients were partitioned into five fixed groups, each serving as the test holdout exactly once. Every group contains at least one atypical-annotated and one normal-annotated patient; the two largest patients (P1: 428 images, P2: 417 images) each anchor a separate group; test-image counts are balanced across groups (315–443 images per group). All five folds used the confirmed production recipe.

The grouped design was motivated in part by the small-sample sensitivity that makes individual patient holdouts structurally unreliable: a single missed cell in a one- or two-annotation test set shifts recall by 50 to 100 percentage points, as seen across the P5-A ceiling folds (see Section 4.1.1 "Fold Level Structural Patterns for P5-A"). Eight patients contributed exactly one annotated cell each (P24, P28, P29, P31, P39, P40, P45, P50); splitting a single annotation across train and holdout is methodologically incoherent, so these patients were always assigned to the training set across all five folds. Their annotations contribute to training in every fold and do not appear in any test split.

Across five groups, mAP50 = 0.872 ± 0.064, mAP50-95 = 0.748 ± 0.079, precision = 0.864 ± 0.022, recall = 0.853 ± 0.087, R_{atypical} = 0.885 ± 0.035, and R_{normal} = 0.821 ± 0.165. The detailed metrics per group are shown in Table 4 in Section 3.2.2 "Per Group Breakdown (P5-B)". Group (G) 3 achieved R_{normal} = 0.981, the highest across all multi-patient evaluation folds in the project.

Additional Phase (P6-A)

A G1–G4 ablation run re-ran four-fold grouped CV on G1–G4 only, with G5 excluded from training, validation, and negative sets. Results are compared against the G1–G4 subset of P5-B (folds 1–4 only, excluding the G5 fold to keep the comparison matched): mAP50 = 0.864 ± 0.041 (+0.000), mAP50-95 = 0.738 ± 0.056 (-0.002), precision = 0.831 ± 0.039 (-0.035), recall = 0.849 ± 0.080 (+0.006), R_{atypical} = 0.897 ± 0.045 (+0.009), R_{normal} = 0.801 ± 0.156 (+0.004). Every delta falls within the cross-fold spread; the largest movement, a 0.035 drop in precision, sits inside the ±0.039 fold standard deviation, while detection recall and both per-class recalls are marginally higher. Training without G5 is therefore feasible with no substantial net performance loss.

This finding raised the question of whether G5 could be withheld entirely as a calibration cohort. That approach was ruled out: to make use of all G5 patients for calibration, inference would need to be run across the full slide set of each G5 patient (roughly 20′000 tiles per patient, approximately 380′000 tiles in total), which is impractical for a single calibration pass. The alternative, withholding all of G5 from training but calibrating on only one of its patients, would discard image information from the remaining G5 patients without benefit. Instead, a single patient was withheld from training: P11, selected because its 17 atypical and 20 normal annotations provide near-equal class balance, the most informative single-patient confidence reference in the corpus. All remaining patients, including the rest of G5, were retained in training. The final deployment model (P6-A in Table 2) was trained on P1–P53 minus P11 (1′832 images, 1′459 atypical, 491 normal, 3.0:1 ratio) using the production recipe unchanged.

Table 2 summarizes the key milestone runs across all phases. The complete experiment record, including corrupted runs and all 11 DoE grid cells, is maintained in `hpc/experiment_log.md` [29]. P3-A per-class recall values in Table 2 are cross-fold means from the tinkering matrix reference configuration (degrees=0, dfl=1.5, freeze=10, label_smoothing=0.0), which is equivalent to the direct P3-A run; folds where the respective class was absent from the held-out patient are excluded from each per-class mean.

Table 2: Milestone run progression summary across all experimental phases. Corpus-size is the total annotated image count for that run. All metrics shown as mean ± std (sample, n-1) across folds except P6-A (single training run, val split only; no held-out test set). Base model is yolo11n throughout except P3-E (yolo11l). CV-codes: S = stratified k-fold, L = leave-one-patient-out (LOPO), CV = grouped cross-validation; prefix = fold/group count.

Run	CV	#imgs	mAP 50-95	Prec.	Recall	R_Atyp	R_Norm
P1-base	5f-S	428	0.551 ± 0.045	0.803 ± 0.044	0.787 ± 0.073	—	—
P2-C	5f-S	417	0.344 ± 0.031	0.880 ± 0.017	0.397 ± 0.022	—	—
P2-H	5f-S	417	0.494 ± 0.166	0.934 ± 0.057	0.537 ± 0.210	—	—
P3-A	8f-L	1′103	0.718 ± 0.137	0.764 ± 0.138	0.840 ± 0.122	0.884 ± 0.170*	0.795 ± 0.164*
P3-B	**8f-L**	**1′103**	**0.741 ± 0.122**	**0.812 ± 0.153**	**0.866 ± 0.131**	**0.892 ± 0.173**	**0.840 ± 0.176**
P3-E	8f-L	1′103	0.745 ± 0.095	0.778 ± 0.134	0.785 ± 0.103	0.875 ± 0.169	0.695 ± 0.144
P4-A	15f-L	1′564	0.793 ± 0.106	0.851 ± 0.097	0.904 ± 0.120	0.913 ± 0.110	0.890 ± 0.194
P4-B	**15f-L**	**1′564**	**0.789 ± 0.118**	**0.826 ± 0.131**	**0.895 ± 0.132**	**0.900 ± 0.174**	**0.882 ± 0.193**
P4-C	15f-L	1′564	0.782 ± 0.124	0.863 ± 0.087	0.893 ± 0.135	0.896 ± 0.172	0.882 ± 0.193
P5-A	35f-L	1′869	0.858 ± 0.120	0.925 ± 0.128	0.951 ± 0.084	0.920 ± 0.129	0.941 ± 0.119
P5-B	**5g-CV**	**1′869**	**0.748 ± 0.079**	**0.864 ± 0.022**	**0.853 ± 0.087**	**0.885 ± 0.035**	**0.821 ± 0.165**
P6-A	val	1′832	0.842	0.906	0.921	0.914	0.929

P1-base established the initial proof of concept on a single patient. P2-C identified the learning rate mismatch that suppressed recall in all $\mathrm{Model}_{\mathrm{P1}}$ runs before the fix; P2-H is the Phase 2 reference after the learning rate and freeze correction. P3-A marks the single largest structural improvement in the project(* = NaN issue, see Measurement Notes). P3-B confirmed the production recipe through the 8-fold tinkering matrix and is the first fully validated configuration (bold). P3-E is a deliberate negative result bounding model-size choices. P4-A documents the corpus expansion gain from P9–P15 under the unchanged degrees=0 recipe, isolating the effect of added data; P4-B is the primary result for the P1–P15 phase because it applies the full P3-B recipe (degrees=5) carried unchanged into all downstream phases, and its marginal deficit to P4-A lies within the fold-to-fold standard deviation. P4-C is the freeze ablation. P5-A is the optimistic upper bound at full corpus scale; P5-B is the primary cross-validated generalization result. P6-A is the final deployment model trained on all 35 annotated patients minus P11; its metrics are validation-split values as no held-out test set exists.

3.2. Quantitative and Qualitative Evaluation

3.2.1. Per-Fold Breakdown (P5-A)

Table 3 presents the complete 35-fold post-hoc evaluation results for P5-A. Each row corresponds to one annotated patient held out from training entirely; all metrics are evaluated on that patient's images exclusively. Post-hoc evaluation was applied uniformly across all 35 folds because Fold 17 (P37 holdout) was killed by the 48-hour SLURM wall-time limit and retrained independently; running the unified post-hoc pass over all folds ensures a consistent result set without mixed evaluation protocols (see Measurement Notes). Folds are listed in numeric order.

Table 3: P5-A per-fold test results (35-fold LOPO post-hoc evaluation, full P1–P53 annotated corpus). Patient / Holdout = sequential fold index / held-out dataset patient. GT Atyp / GT Norm = ground-truth cell instance counts in the held-out test split. "—" indicates the class was absent from that test split and is excluded from the per-class mean. R_Atyp mean over n=20 folds; R_Norm mean over n=29 folds.

Fold / Holdout	GT Atyp	GT Norm	mAP 50-95	mAP50	Prec.	Recall	R_Atyp	R_Norm
1 / P1	482	2	0.610	0.813	0.577	0.973	0.946	1.000
2 / P10	5	20	0.909	0.991	0.987	0.906	1.000	0.812
3 / P11	17	20	0.863	0.964	0.968	0.887	0.941	0.832
4 / P12	133	0	0.832	0.984	0.962	0.952	0.952	—
5 / P13	0	7	0.920	0.995	1.000	1.000	—	1.000
6 / P14	5	3	0.933	0.995	0.978	1.000	1.000	1.000
7 / P15	0	68	0.949	0.991	0.984	0.985	—	0.985
8 / P16	22	1	0.834	0.994	1.000	0.977	0.955	1.000
9 / P17	69	0	0.832	0.992	0.945	0.994	0.994	—
10 / P18	81	1	0.948	0.993	0.934	0.946	0.893	1.000
11 / P2	449	6	0.600	0.716	0.721	0.712	0.924	0.500
12 / P24	1	0	0.796	0.995	1.000	1.000	1.000	—
13 / P28	1	0	0.995	0.995	1.000	1.000	1.000	—
14 / P29	1	0	0.896	0.995	1.000	1.000	1.000	—
15 / P3	60	4	0.736	0.906	0.785	0.931	0.911	0.951
16 / P31	0	1	0.995	0.995	1.000	1.000	—	1.000
17 / P37	0	2	0.896	0.995	1.000	1.000	—	1.000
18 / P39	0	1	0.896	0.995	1.000	1.000	—	1.000
19 / P4	122	6	0.675	0.806	0.754	0.796	0.926	0.667
20 / P40	0	1	0.995	0.995	1.000	1.000	—	1.000
21 / P43	0	3	0.940	0.995	1.000	1.000	—	1.000

Patient / Holdout	GT Atyp	GT Norm	mAP 50-95	mAP50	Prec.	Recall	R_Atyp	R_Norm
22 / P44	0	2	0.921	0.995	1.000	1.000	—	1.000
23 / P45	0	1	0.995	0.995	1.000	1.000	—	1.000
24 / P46	0	4	0.967	0.995	1.000	1.000	—	1.000
25 / P47	0	3	0.918	0.995	1.000	1.000	—	1.000
26 / P48	7	0	0.857	0.995	1.000	1.000	1.000	—
27 / P5	3	18	0.652	0.760	0.595	0.740	0.667	0.814
28 / P50	0	1	0.995	0.995	1.000	1.000	—	1.000
29 / P51	5	90	0.700	0.765	0.740	0.900	0.800	1.000
30 / P52	1	4	0.945	0.995	0.971	1.000	1.000	1.000
31 / P53	0	5	0.952	0.995	1.000	1.000	—	1.000
32 / P6	0	6	0.846	0.995	1.000	1.000	—	1.000
33 / P7	8	12	0.809	0.975	0.929	0.917	1.000	0.833
34 / P8	0	24	0.816	0.986	0.914	0.958	—	0.958
35 / P9	4	195	0.600	0.622	0.635	0.722	0.500	0.944
Mean			**0.858**	**0.948**	**0.925**	**0.951**	**0.920** (n=20)	**0.941** (n=29)

P1 R_{normal} = 1.000 based on 2 normal ground-truth instances and is not a stable estimate. P12 has 133 atypical and 0 normal ground-truth instances; R_{normal} is absent from its test set.

Nineteen folds reported mAP50 at the ceiling value of 0.995 (folds 5, 6, 12–14, 16–18, 20–26, 28, 30–32). The majority hold out patients from the P16–P53 acquisition batch who contributed one to three annotated images each; folds 5, 6, and 32 (holdouts P13, P14, and P6) also reach the ceiling for the same structural reason. On a one or two-image test set, detecting the single annotated cell forces recall to its maximum independent of model quality. mAP50 reaching the 0.995 ceiling on the same fold is a weaker but real statement: it requires that the model produced no false positive with a confidence higher than the true detection on that image, since any such high-confidence false alarm would depress precision early in the PR curve and lower the integrated AP. The ceiling therefore confirms that the model suppressed high-confidence background artefacts on that specific image, but a single-image fold still carries no cross-patient generalization signal. These folds confirm that the model produces no catastrophic failures on any individual patient from the expanded corpus. The six folds with mAP50 below 0.820 (folds 1, 11, 19, 27, 29, and 35; holdouts P1, P2, P4, P5, P51, and P9) represent the meaningful lower tail of the performance distribution at full corpus scale; their per-fold patterns are examined in Section 4.1.1 “Fold Level Structural Patterns for P5-A”.

3.2.2. Per-Group Breakdown (P5-B)

Table 4 presents the complete per-group test results for P5-B. Each row corresponds to one held-out patient group; all metrics are evaluated exclusively on images from patients in that group. The holdout column identifies the constituent patients; G5 contains the six large-corpus patients from P4 and P5 plus all 13 remaining small-corpus patients from P16–P53. See Section 3.2.2 "Group-Level Structural Patterns for P5-B" for detailed analysis.

Table 4: P5-B grouped 5-fold CV split composition and test results (35 annotated patients, P1–P53 corpus). Train = annotated (positive) images; each fold additionally receives 3′773-4′595 negative tiles. Atyp/Norm = ground-truth label counts in the test split. All five groups have both classes present.

Group	Holdout patients	Train	Val	Test	Atyp	Norm	mAP 50-95	mAP50	Prec.	Recall	R_Atyp	R_Norm
G1	P1, P13, P31	1219	214	436	482	10	0.649	0.766	0.850	0.746	0.904	0.588
G2	P2, P8, P37	1213	213	443	449	32	0.679	0.858	0.839	0.773	0.836	0.710
G3	P9, P12, P53	1315	231	323	137	200	0.813	0.913	0.882	0.932	0.884	0.981
G4	P3, P6, P15–P18, P48	1322	232	315	239	80	0.821	0.918	0.892	0.919	0.928	0.909
G5	*	1289	228	352	169	189	0.777	0.905	0.854	0.896	0.873	0.918
Mean ± std							**0.748 ± 0.079**	**0.872 ± 0.064**	**0.864 ± 0.022**	**0.853 ± 0.087**	**0.885 ± 0.035**	**0.821 ± 0.165**

* P4, P5, P7, P10–P11, P14, P24, P28–P29, P39–P40, P43–P47, P50–P52

3.3. Evaluation Visualization

Training convergence for P5-B is shown in Figure 13 for a representative fold (Fold 3, G3 holdout: P9, P12, P53; mAP50 = 0.913, recall = 0.932). The training and validation loss curves converge within approximately 150 epochs on the majority of folds, with YOLO's early-stopping criterion (patience = 50) triggering at the validation mAP50 plateau.

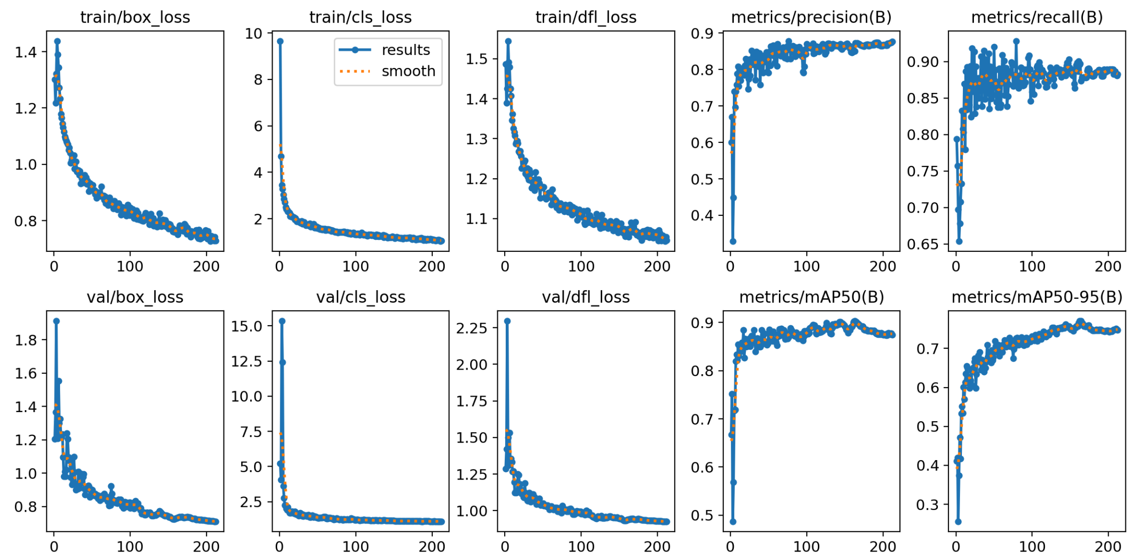

Figure 13: Training convergence for P5-B Fold 3 (G3 holdout: P9, P12, P53). Panels from top left: box loss (train/val), classification loss (train/val), and distribution focal loss for training and validation splits; precision, recall, mAP50, and mAP50-95 across epochs.

The normalized confusion matrix for the same fold is shown in Figure 14. Column normalization is applied because the atypical and normal test-set sizes differ, making raw counts incomparable across classes; column-normalized values represent the fraction of each ground-truth class assigned to each predicted class. The main diagonal records 0.96 for atypical and 0.88 for normal, indicating the fraction of each class correctly detected. Cross-class misclassifications between atypical and normal are negligible at 0.01. The dominant off-diagonal entry is 0.91 in the background-to-atypical cell, meaning that detections not matching any ground-truth cell are predominantly assigned to the atypical class; these are FP, and their clinical handling is discussed in Section 4.3 "Error Analysis". The raw numbers are visible in Figure 15.

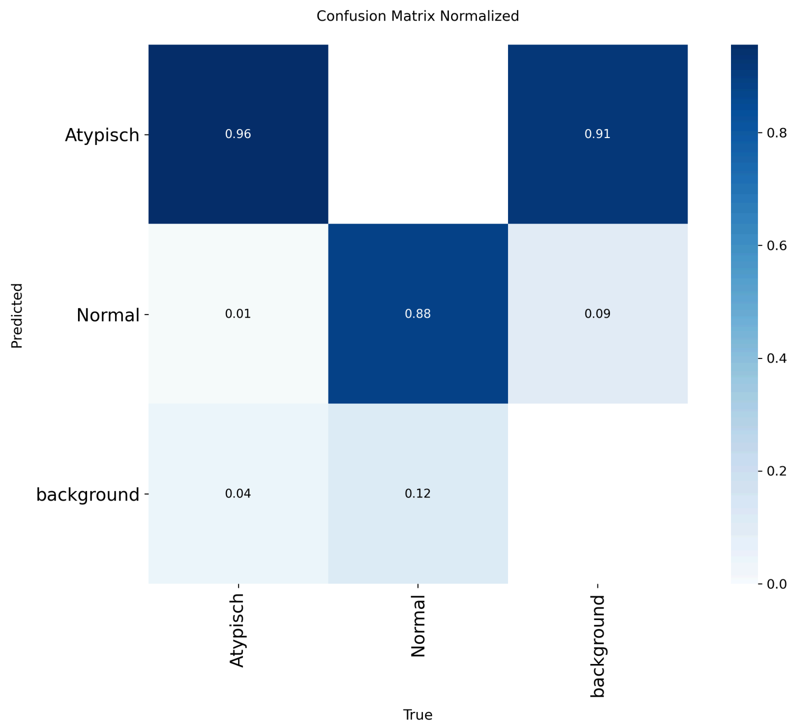

Figure 14: Normalized confusion matrix for P5-B Fold 3 (G3 holdout: P9, P12, P53, evaluated on the val split). Rows are predicted classes; columns are ground-truth classes. Values are row-normalized fractions. The main diagonal entries are 0.96 (atypical) and 0.88 (normal); cross-class error is 0.01; background-to-atypical is 0.91.

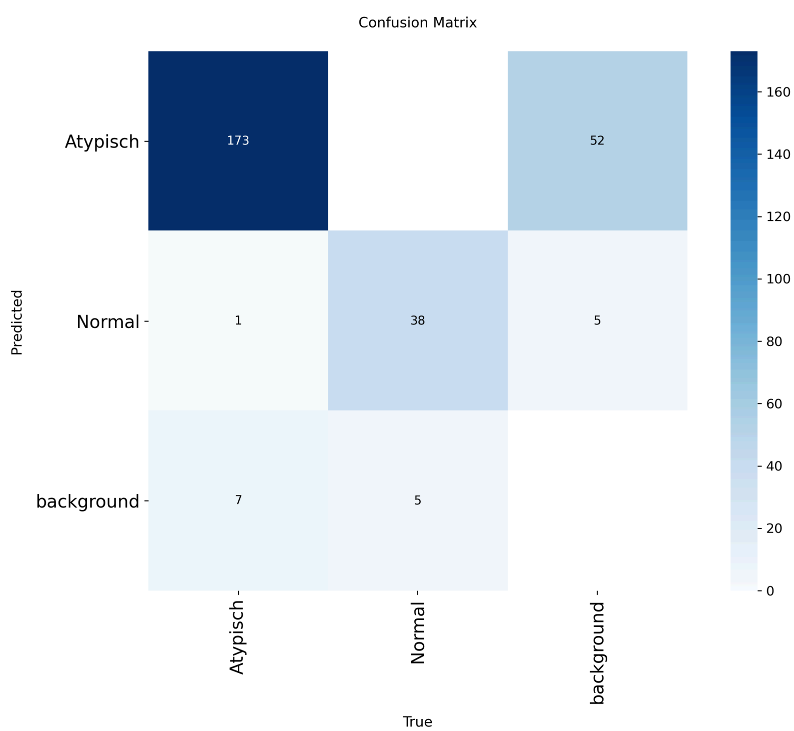

Figure 15: Confusion matrix for P5-B raw numbers.

3.3.1. Phase 5–6 Performance Comparison

Figure 16 shows per-group R_{atypical} and R_{normal} for all five P5-B folds alongside the cross-fold means.

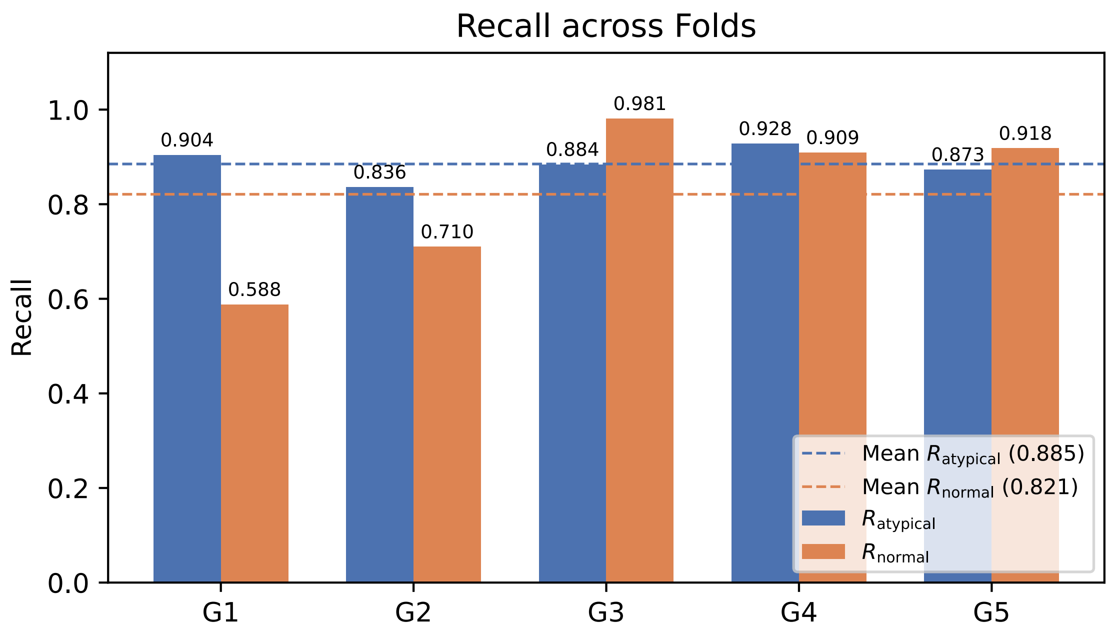

Figure 16: P5-B per-group recall by class. Bars show R_{atypical} (blue) and R_{normal} (orange) for each held-out patient group. Dashed horizontal lines mark the cross-fold means ($R_{\text{atypical}} = 0.885$, $R_{\text{normal}} = 0.821$).

Figure 17 compares all six evaluation metrics between P5-A and P5-B. Error bars show one standard deviation across folds for both evaluations; R_Atyp and R_Norm for P5-A are computed over the subset of folds where the respective class was present in the test set.

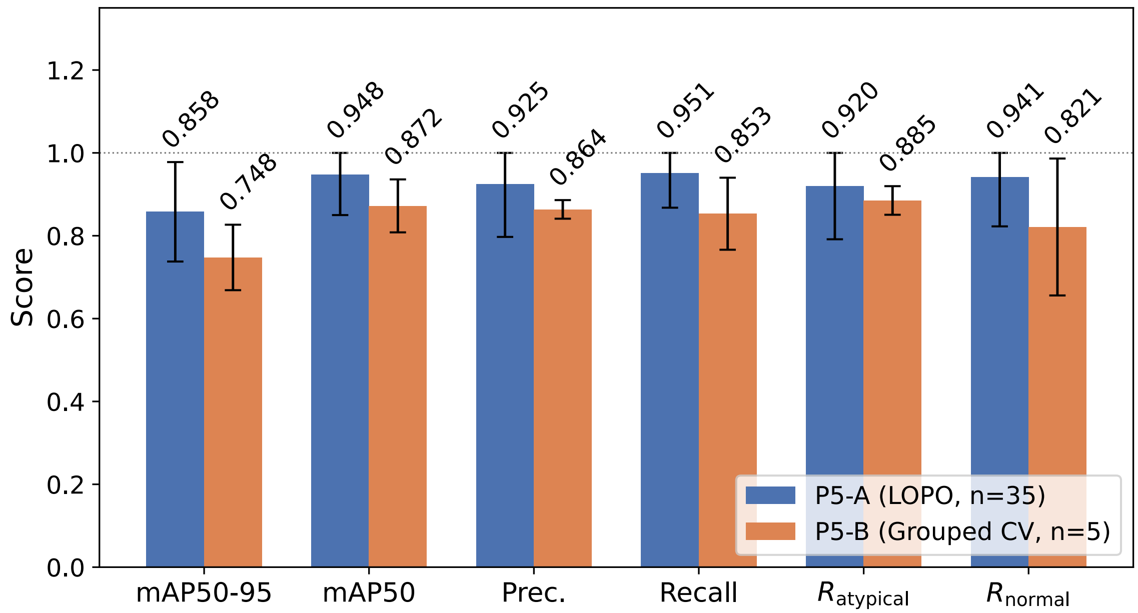

Figure 17: Metric comparison between P5-A (35-fold LOPO) and P5-B (grouped 5-fold CV). Error bars represent ±1 standard deviation across folds. R_Atyp and R_Norm for P5-A are computed over the 20 and 29 folds respectively where the class was present in the test set.

Figure 18 shows mAP50-95, Precision, and Recall for P5-A (35-fold LOPO, test split, n_atypical=20, n_normal=29), P5-B (5-group CV, test split), and P6-A (final deployment model, validation split). P6-A has no held-out test set as it was trained on all annotated patients minus P11.

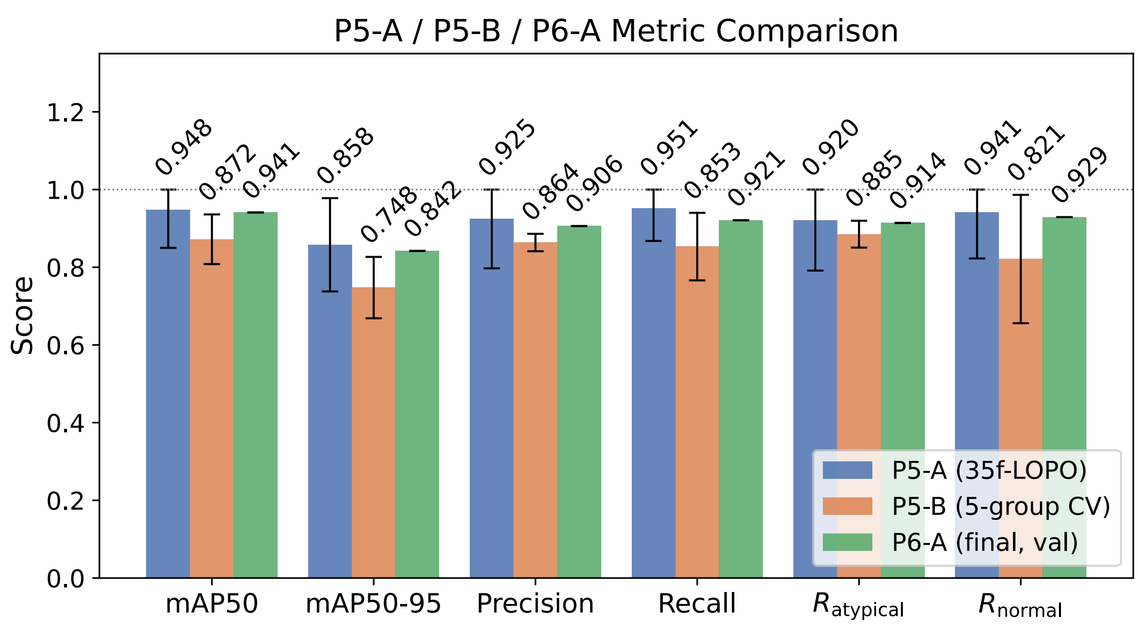

Figure 18: Performance comparison across P5-A , P5-B, and P6-A. Error bars show std across folds, clipped at 1.0.

3.3.2. Final Model Training Visualization

Training convergence for the P6-A deployment model is shown in Figure 19. The model trained on 1′832 images (P1–P53 minus P11) and converged at epoch 607 (early stopping, patience=50). It shows a clean convergence towards higher epochs. No overfitting present.

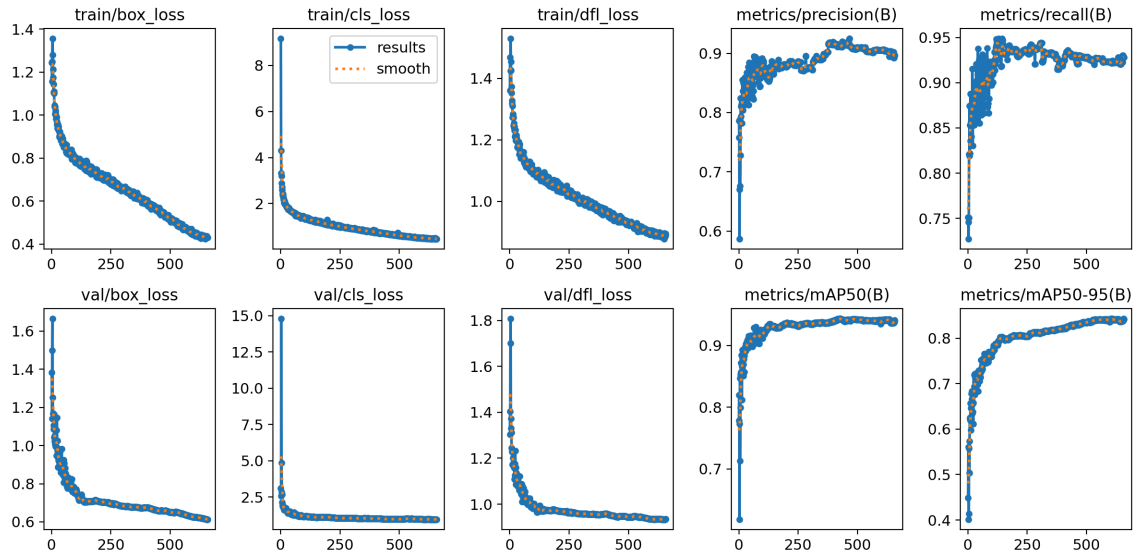

Figure 19: Training curves for the P6-A final deployment model.

The normalized confusion matrix at best.pt is shown in Figure 20. Evaluated on the validation split (15% of training data). The raw numbers are visible in Figure 21.

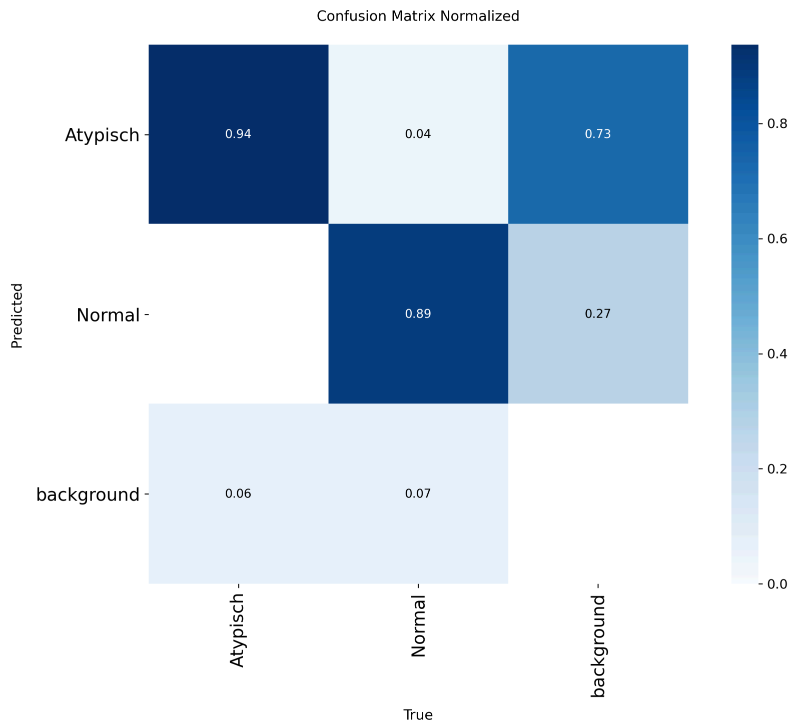

Figure 20: Normalized confusion matrix for P6-A (best.pt, evaluated on the val split). Rows are predicted classes; columns are ground-truth classes.

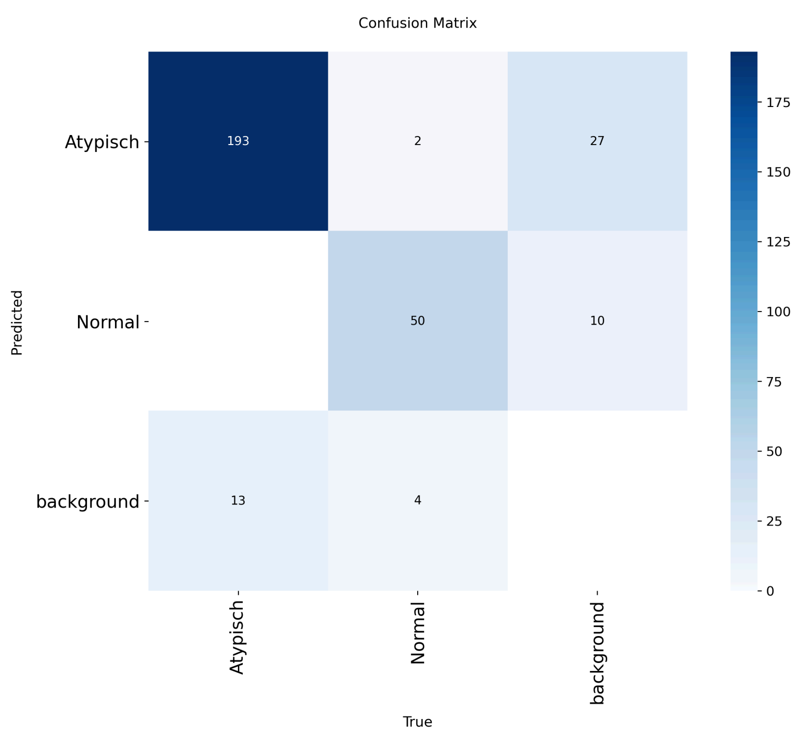

Figure 21: Confusion matrix for P6-A raw numbers.

3.4. Real World Usage

3.4.1. Confidence Threshold Calibration

The final deployment model (P6-A) was applied out of sample to the complete P11 tile set, and the resulting confidence distribution is shown in Figure 22. P11 was selected for this role because the near-equal class balance of 17 atypical and 20 normal annotations makes it the most informative single-patient calibration reference available without retraining; any other annotated patient would have contributed to P6-A's training set and would produce in-sample confidence scores. The histogram is delivered to USZ so the haematologist can select an operating threshold against the clinic's own recall and review-burden tradeoff, rather than inheriting a single hard-coded engineering value.

P11 is, however, an insufficient calibration cohort for placing that threshold with statistical confidence. With only 37 annotated cells across approximately 12′000 tiles, the model produces a small number of high-confidence true-positive detections against a large bulk of low-confidence background artefacts. The histogram does exhibit the expected bimodal structure, a broad low-confidence cluster from background and a narrower high-confidence peak from true detections, but the true-positive peak rests on too few detections for the separation point to be located reliably. A threshold drawn from this distribution is directionally informative but not a precise operating recommendation.

The practical constraint behind this limitation is data scarcity. P1 (482 atypical, 2 normal) and P2 (449 atypical, 6 normal) each carry roughly twenty-five times the annotated cell count of P11. Withholding either from the 35-patient training corpus specifically for calibration would have removed the two largest single contributors to atypical-class coverage from training, and that tradeoff was not acceptable at this corpus size. P11′s modest annotation count made the scarcity cost acceptable; its class balance was a secondary benefit.

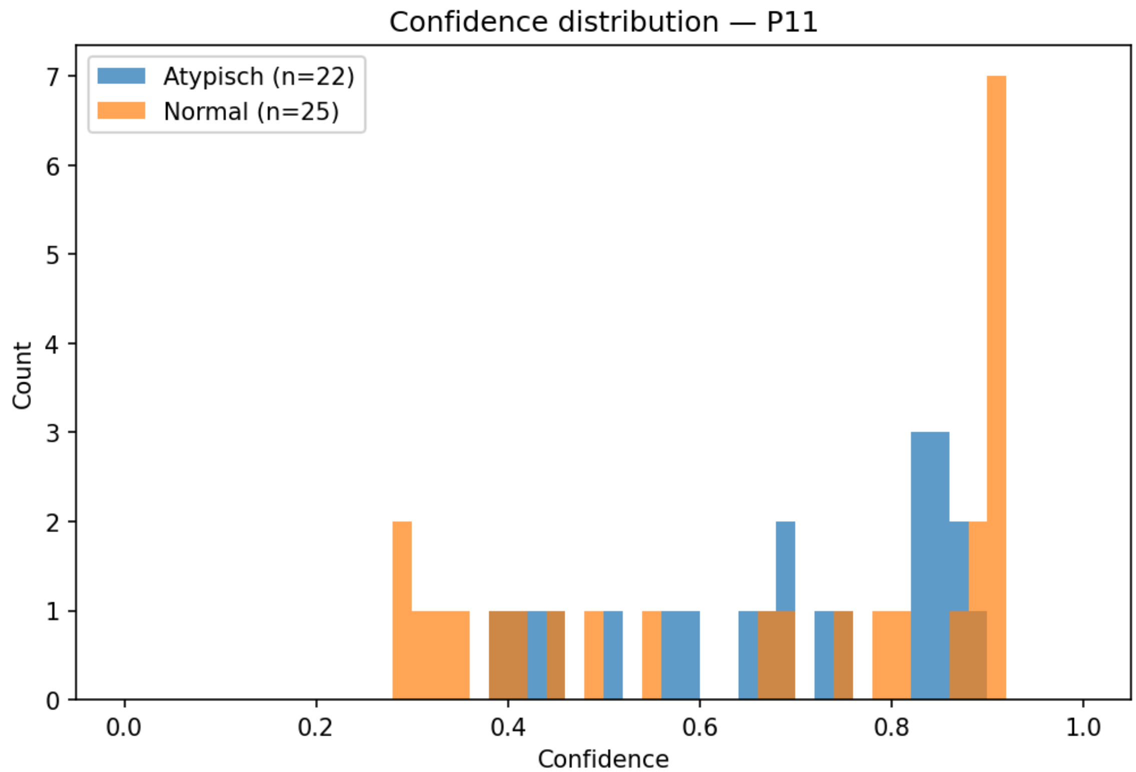

Figure 22: Confidence distribution of the final deployment model (P6-A) across the complete P11 tile set (~12′000 tiles). The counts (n=22 atypical, n=25 normal) exceed the 17 atypical and 20 normal annotations because they include FP. Bins containing both classes appear dark orange, as the semi-transparent blue and orange bars overlap.

4. Discussion

4.1. Interpretation of Results

To our knowledge, this is the first deep learning pipeline for the specific diagnostic task of distinguishing atypical from normal mast cells in bone marrow aspirates, validated under patient-held-out cross-validation and aligned to the WHO ≥25% minor criterion for systemic mastocytosis. The systematic review by Ghete et al. [9] of DL-based bone marrow cell classification from 2019 to 2024 documents work on broad cell-type classification but none targeting the atypical-versus-normal mast cell distinction that the WHO ratio depends on, and few studies validating on patient-held-out splits at all. The work is also delivered, not just measured: the MastCellDetector desktop application [28] runs the final model offline at USZ and reports the atypical fraction directly against the ≥25% threshold, turning the pipeline into a tool that participates in the diagnostic decision rather than a benchmark number.

The primary generalization estimate is the P5-B grouped 5-fold cross-validation: mAP50-95 = 0.748, precision = 0.864, recall = 0.853, R_{atypical} = 0.885, and R_{normal} = 0.821 across 35 annotated patients, with no image from a held-out patient ever appearing in training within its fold. These are conservative, deployment-relevant numbers: the cross-fold standard deviation reflects genuine inter-patient biological variability rather than random-split noise, and the test groups (315 to 443 images each) are large enough to absorb individual missed detections without catastrophic recall swings. The 35-fold LOPO run (P5-A: R_{atypical} = 0.920, R_{normal} = 0.941) is reported as an optimistic upper bound only. Its per-class means are inflated by 19 folds at the mAP50 ceiling of 0.995, where a single-image test set guarantees perfect recall once its one annotated cell is detected, independent of any generalization capacity. P5-B is the number a clinician should trust; P5-A bounds the best case.

The final deployment model (P6-A), trained on all 35 annotated patients minus P11, achieves R_{atypical} = 0.914 and R_{normal} = 0.929 on its best validation split, consistent with the P5-B grouped-CV estimate and confirming that the production recipe generalizes cleanly to the expanded training set. These are in-sample validation metrics, not held-out test results, but their alignment with P5-B is reassuring evidence against overfitting at deployment scale.

R_{normal} is the load-bearing metric for this task, because undercounting normal cells inflates the atypical fraction and can push a borderline patient across the ≥25% threshold. The single largest improvement in R_{normal} at any point in the project came from corpus expansion, specifically the addition of patient P9 with its 195 normal annotations, and it exceeded the gain from any individual hyperparameter change. This is not an incidental observation. The Phase 2 design-of-experiment grid varied false-positive oversampling and background-tile ratio across the full composition space of a fixed corpus and found no statistically distinguishable difference: the spread across all configurations was smaller than the cross-fold standard deviation within any single run. The controlled conclusion is that when class imbalance is structural rather than accidental, that is, when the minority class is rare because the data is sampled from disease-positive patients in whom that morphology is genuinely uncommon, reweighting and resampling the existing data cannot substitute for collecting more biologically diverse patients. For rare-event medical imaging this is a transferable lesson: corpus expansion, not hyperparameter tuning, is the primary lever for rare-class recall under structural imbalance. The trajectory bears this out, with R_{normal} reaching 0.981 on the best multi-patient fold (G3) once the corpus had acquired sufficient normal-class coverage.

The automated hyperparameter tuner (P3-C) reinforces this point from a different angle. YOLO's built-in tuner, run for 30 iterations on a single atypical-dominant holdout patient (P4), converged on degrees=0.0 and dfl=2.47, both parameters that the cross-validated tinkering matrix had identified as harmful to R_{normal}. The tuner behaved rationally given its objective, mAP50-95 on a P4 holdout, but that objective rewards atypical-class performance at the expense of normal-class generalization. This finding generalizes: for rare-class tasks where class imbalance is structural, standard automated hyperparameter search requires multi-patient cross-validation to prevent the dominant class from steering the search. Single-fold tuning is blind to the recall balance that the diagnostic criterion depends on. The production recipe retained from the tinkering matrix was not a conservative choice; it was the only choice that a cross-validated evaluation could identify as correct.

A second methodological contribution is the bootstrapped, model-assisted annotation loop. The first patient was annotated by hand to train an initial detector; each subsequent model generation then served as the CVAT inference backend for the next batch of patients, and the false positives the annotator confirmed at each round became curated hard negatives for the following training cycle. This coupling of annotation and model development is, to our knowledge, not documented for mast cell detection, and it is what made a 53-patient corpus feasible within a single thesis: the Ghete et al. review notes that most published studies apply models to pre-existing datasets rather than building the annotation pipeline alongside the model. Here the clinical partner's annotation capacity was extended by each model generation, and each generation's hardest mistakes were fed forward as training signal rather than discarded.

Finally, the threshold question is handled honestly. Rather than shipping a single hard-coded confidence value, the project withheld patient P11 from all training as a near-balanced calibration cohort, ran the final model over its full tile set out of sample, and delivered the resulting confidence histogram to USZ so the hospital can choose its own operating point against its own recall and review-burden tradeoff. A deployed tool that hands the institution an out-of-sample calibration reference, rather than an in-sample number, is the appropriate form for a clinical decision that is ultimately the haematologist's to make.

G3, G4, and G5 each achieve recall above 0.89, consistent with the P4-B 15-patient LOPO mean of 0.895. This stability confirms that expanding from 15 to 35 patients did not introduce distributional drift: the model trained on the broader corpus generalizes at least as well on the earlier patient cohort as the Phase 4 model did.

The grouped CV design is not equivalent to LOPO, but it preserves the property that matters: no image from a holdout patient appears in its fold's training set. It was also the tractable choice. Five folds cost roughly 36 hours at MAX_PARALLEL = 2, fitting the 96-hour HPC allocation that made a full 35-fold LOPO impractical to run as the primary protocol. Grouped patient CV is therefore the deployable approximation of LOPO at this corpus scale.

4.1.1. Fold-Level Structural Patterns for P5-A

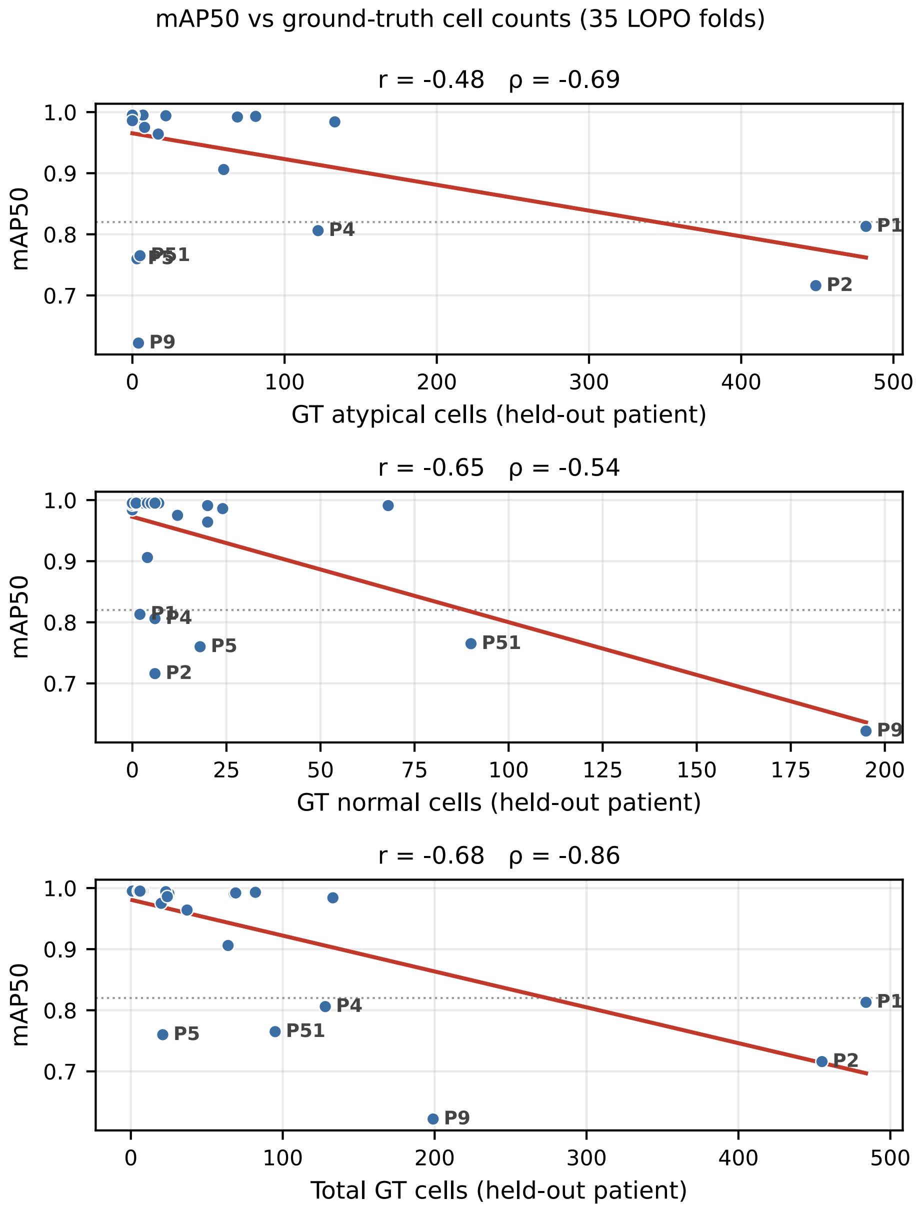

Figure 23: Correlation of P5-A 35-fold LOPO: mAP50 against ground-truth cell counts. Marked points are below the threshold of 0.820.

At first glance, Table 3 suggests a negative correlation between the atypical ground-truth count and mAP50: the three folds with the largest atypical burden (P1, P2, P4) all fall below the 0.820 threshold, and the least-squares fit in Figure 23 mirrors this with a downward slope (Pearson r = −0.48). The correlation is real, but confirming a correlation and confirming a cause are two different things. The slope says that folds with more atypical cells tend to score lower; it does not say that atypical cells are harder to detect, which is the interpretation the downward trend invites. That interpretation would require low atypical recall, and recall is in fact high on exactly these folds (R_Atyp ≥ 0.92), so the correlation does not act through the mechanism it appears to suggest. Several observations confirm this. The atypical count is the weakest of the three predictors; the normal count (r = −0.65) and especially the total ground-truth count (r = −0.68, Spearman ρ = −0.86) track mAP50 more closely, indicating that fold difficulty is governed by overall test-set size and class balance rather than by atypical burden. The decisive counter-example is P12: with 133 atypical cells, the third-highest count in the study and more than P4′s 122, it scores mAP50 = 0.984, while the comparably high-atypical folds P17 (69) and P18 (81) both exceed 0.99. P4 and P12 are matched in atypical burden, and in per-image cell density, yet land on opposite sides of the threshold, and the difference is precision, not count. P4′s precision is 0.754 against P12′s 0.962, and all three sub-threshold high-atypical folds are precision-limited (P1 0.577, P2 0.721, P4 0.754) while retaining high atypical recall (≥ 0.92). Their low scores reflect reduced precision rather than any difficulty caused by the atypical count itself; for P1, the combination of very high recall (0.973) and low precision (0.577) is consistent with its earlier annotation regime, in which unlabelled true cells are scored as false positives.

Patient 9 is the weakest informative fold.

Patient 9 contributes 4 atypical and 195 normal instances to the test set. A single missed atypical cell moves R_{atypical} by 25 percentage points; P5-A recorded R_{atypical} = 0.500 (2 of 4 detected). The resulting mAP50 of 0.622 is the lowest across all 35 folds. Patient 9′s extreme class asymmetry (4 atypical against 195 normal) makes it the hardest fold by construction: the model trains on a broadly atypical-dominant corpus but must classify correctly in a patient where normal cells outnumber atypical by nearly 50:1. The high R_{normal} of 0.944 on the same fold confirms that the model handles the normal-dominant distribution correctly when Patient 9 is withheld; the weakness is specific to the four rare atypical instances.

Patient 1 recovered from its Phase 3–4 structural weakness.

The Patient 1 holdout fold, which produced mAP50 = 0.584 under the 14-patient P4-B training set, improved to mAP50 = 0.813 under the 34-patient P5-A training set. The annotation-regime gap between Patient 1′s original `Pos_neg` directory structure and the CVAT-assisted P2–P53 annotations remains in place, but the expanded corpus provides enough additional atypical-dominant training material to partially compensate for the distributional mismatch. R_{atypical} = 0.946 and R_{normal} = 1.000 (on 2 normal instances) are both higher than in any prior phase under this holdout.

Patient 2 and Patient 5 show small-sample minority-class fragility.

The Patient 2 holdout (449 atypical, 6 normal) produced R_{normal} = 0.500 (3 of 6 detected); a single additional missed normal cell would reduce this to 0.333. Patient 5 (3 atypical, 18 normal) recorded R_{atypical} = 0.667 (2 of 3 detected); one missed atypical shifts the value by 33 percentage points. Both figures are dominated by small-sample noise rather than systematic model error. P5′s R_{normal} = 0.814 and P2′s R_{atypical} = 0.924 confirm that each patient’s majority class is detected reliably; the fragility is confined to the minority class within each specific test set.

Ceiling folds are structurally uninformative.

The 19 folds at mAP50 = 0.995 are patients who contributed one to three annotated images each, predominantly from the P16–P53 acquisition batch but including three early patients (P6, P13, P14). Perfect or near-perfect recall on a one-image test set is mathematically guaranteed when the single annotated cell is detected; these values confirm the absence of catastrophic failures but carry no information about cross-patient generalization capacity. Per-fold recall values throughout Table 3 are informative as lower bounds (for structurally fragile small-sample folds) and as ceiling observations rather than as stable point estimates.

4.1.2. Group-Level Structural Patterns for P5-B

The grouped evaluation shows the same surface pattern as the per-patient folds: the test groups with the largest atypical counts also record the lowest mAP50. With only five groups this ordering is not quantified as a correlation, but the same caution applies as for P5-A: a correlation and a cause are not the same thing. Even read as a trend, "lower mAP on atypical-heavy groups" does not mean atypical cells are harder to detect, since that would require low atypical recall, and atypical recall is high on exactly these groups. The grouped data does not support that reading.

The low scores are confined to the two SM-dominant groups. The two lowest-scoring groups are G1 (mAP50 0.766) and G2 (mAP50 0.858), and these are precisely the groups that withhold the pure-SM patients P1 and P2. The atypical content of each group originates almost entirely from that single patient: G1's 482 atypical cells are P1's, and G2's 449 are P2's, since the other holdouts in those groups (P13, P31, P8, P37) contribute no atypical instances. The group-level pattern therefore restates the per-patient behaviour of P1 and P2 rather than revealing a separate effect.

Atypical recall remains high in every group, including the lowest-scoring ones. Across the five groups R_{atypical} ranges from 0.836 to 0.928 (mean 0.885), and in the lowest-mAP group G1 it is 0.904. If atypical cells were difficult to detect, atypical recall would fall in the atypical-heavy groups; instead it stays high throughout. The model reliably localises the atypical cells that are present.

In the low-scoring groups the weakness is the normal class, not the atypical one. G1's reduced mAP is driven by its normal-class recall (R_{normal} = 0.588) rather than by atypical detection (R_{atypical} = 0.904); the overall recall of 0.746 is the mean of these two per-class values. G1 contains only 10 normal instances, so missing roughly four of them moves R_{normal} by tens of percentage points. This is the same small-sample minority-class fragility documented for P5-A, expressed here on the normal class because the SM-dominant groups are normal-sparse.

A high atypical count does not by itself depress mAP. Group G3 contains the high-atypical patient P12 (133 atypical cells) yet scores mAP50 = 0.913, in line with the per-patient finding that atypical burden does not determine fold difficulty. The apparent association in G1 and G2 reflects their SM-dominant, normal-sparse composition, not the number of atypical cells as such.

Taken together, the grouped results agree with the per-patient evaluation. The atypical class is detected reliably across all groups; the mAP variation tracks the SM-dominant composition of G1 and G2 and the fragility of their sparse normal class, rather than any difficulty in detecting atypical cells.

4.1.3. Measurement Notes

A per-class recall extraction issue affected all LOPO runs through Phase 4. The utility function indexed Ultralytics' per-class result arrays by position rather than by the `ap_class_index` field. When a holdout patient contained only one class and the model produced predictions for that class only, Ultralytics returned a one-element rather than a two-element result array, causing positional indexing to assign the single element to R_{atypical} and report R_{normal} as NaN. This affected Fold 5 (P13 holdout) in all Phase 3 and Phase 4 runs; the true R_{normal} for that fold is 1.000, confirmed from the per-fold confusion matrix. Affected folds are excluded from per-class cross-fold means and clearly marked in all fold tables.

For the P5-A post-hoc evaluation, the bug was fixed by examining per-patient ground-truth label files before calling `val()`: the label-derived GT class counts provided a reliable class-index lookup independent of Ultralytics' internal `ap_class_index` field. This fix allowed the 35-fold P5-A results to be extracted correctly without per-fold manual inspection; folds where only one class had ground-truth instances were routed to the appropriate class slot by GT count rather than by position. The aggregate metrics mAP50 and overall Recall are computed internally by Ultralytics and are unaffected by this issue in all phases.

- **RQ1** asked whether the pipeline reliably distinguishes typical from atypical mast cell phenotypes to support the WHO ≥25% criterion. Yes. Under patient-held-out cross-validation the pipeline achieves R_{atypical} = 0.885 and R_{normal} = 0.821 (P5-B, conservative estimate); the final deployment model reports R_{atypical} = 0.914 and R_{normal} = 0.929 on its best validation split (P6-A). Both classes that enter the atypical-to-total ratio are counted with reliable recall, which is the necessary condition for computing a trustworthy fraction. The MastCellDetector application directly operationalizes the answer: its Statistics tab aggregates per-tile detections into a patient-level atypical fraction and displays it against the ≥25% threshold, mapping the pipeline's output onto the diagnostic question without further manual computation.
- **RQ2** asked which performance metrics are most critical for clinical reliability. The answer is per-class recall tracked independently for both morphological classes. A false negative, a missed mast cell, biases the atypical fraction at the diagnostic decision boundary; R_{normal} is the harder of the two because normal is the minority class in the training corpus. mAP50 serves as the run-to-run comparison summary. F1-score is not used as a primary metric because it treats false positives and false negatives symmetrically, whereas this task is explicitly asymmetric: false positives are recoverable at the manual review step, while false negatives silently distort the WHO ratio. Precision is monitored but secondary for the same reason.
- **RQ3** asked whether the pipeline achieves rare-event detection without generating an unmanageable number of false positives. The evidence supports a qualified yes, conditional on the operating threshold. Mast cells occupy only a small fraction of the tiles in a full slide scan (~20′000 tiles per patient), and the confirmed-false-positive hard-negative loop suppresses the most common background error modes across successive annotation generations. Under the conservative P5-B estimate the rare cells are detected reliably (R_{atypical} = 0.885, R_{normal} = 0.821), and on the representative G3 fold the confusion matrix records main-diagonal recall of 0.96 (atypical) and 0.88 (normal) with cross-class confusion at 0.01, so the residual error is dominated by background false alarms rather than atypical-versus-normal mix-ups. That false-alarm load is not a fixed property of the model but a function of the confidence threshold: the matrix is computed at the Ultralytics default of conf = 0.25 [25], while the out-of-sample P11 histogram (Figure 22) separates the low-confidence background bulk from the high-confidence true-positive peak, so raising the threshold trades a small amount of recall for a large reduction in surviving false positives. The figure that would settle RQ2 outright needs to be computed in a future project if the necessity is given by the USZ: the absolute error between the predicted atypical fraction and the ground-truth fraction at a fixed operating point, measured on a dedicated calibration cohort larger than the 37 annotated cells of P11. The defensible answer is therefore that the pipeline detects rare events reliably and keeps false positives recoverable within the MastCellDetector review workflow, while the precise false-positive rate at the deployed threshold remains to be quantified on a larger calibration set, a deployment-evaluation step identified in Section 5.3..

4.2. Clinical Relevance

The clinical value of this pipeline follows directly from the diagnostic question it is aligned to. The WHO minor criterion for SM is met when at least ≥25% of mast cells in the aspirate are atypical [1], so the quantity that matters is a ratio of two counts, not a single detection score. The pipeline is built around that ratio: it counts atypical and normal cells separately and reports the atypical fraction against the threshold.

This is why the error asymmetry is treated the way it is. A false negative, a missed mast cell, biases the ratio and can move a borderline patient across the ≥25% line and delay an SM diagnosis, whereas a false positive is recoverable at the manual review step and only matters near the threshold. The evaluation therefore prioritizes recall, and normal-class recall in particular, because undercounting normal cells inflates the atypical fraction toward a false-positive classification. Optimizing for aggregate mAP, as adjacent white-blood-cell studies typically do [10], would obscure exactly the per-class behaviour the diagnosis depends on.

The bridge from model to clinic is the MastCellDetector desktop application [28]. It runs entirely offline, which satisfies the network restrictions on hospital clinical workstations; it streams label files to disk during inference so a long run is recoverable; and it lets the haematologist filter tens of thousands of tiles down to the cases that need review (with detections, low confidence, single-class) instead of scrolling through a whole slide. Its Statistics tab produces the patient-level atypical fraction against the ≥25% threshold as an exportable summary for clinical documentation. The annotation editor closes the loop: corrections made during clinical use are written back as label files and confirmed negatives, feeding the next training generation. Rather than asking the clinic to trust a fixed threshold, the project delivered an out-of-sample confidence calibration from the withheld patient P11 so USZ sets its own operating point. The intended role is assistive: the tool does the counting and surfaces the candidates, and the haematologist makes the diagnosis.

4.3. Error Analysis

The pipeline's errors cluster into five understood modes, none of which is a general detection failure.

Minority-class small-sample fragility. The largest visible errors are concentrated in folds where the held-out patient contributes only a handful of minority-class cells. In P5-A, P9 (4 atypical against 195 normal) records $R_{\text{atypical}} = 0.500$ because two missed cells out of four move the metric by 50 points, and P2 (6 normal) records $R_{\text{normal}} = 0.500$ on three missed cells. These are properties of the test set, not of the model: each patient's majority class is detected reliably on the same fold (P9 $R_{\text{normal}} = 0.944$, P2 $R_{\text{atypical}} = 0.924$). Such folds are informative as lower bounds, not as point estimates, and are handled in the reported means by restricting per-class averages to folds with enough minority-class instances.

The Patient 1 annotation-regime outlier. Patient 1 was annotated under the original Pos_neg directory structure before the CVAT-assisted workflow, and its holdout fold was the weakest in Phase 3 and Phase 4 (mAP50 = 0.584 under the 14-patient training set). The expanded corpus partly compensated for this distribution mismatch (Patient 1 mAP50 recovered to 0.813 under the 34-patient P5-A training set), but the annotation-regime difference remains a known source of error and is the main driver of G1′s depressed normal recall in P5-B.

The per-class recall measurement bug. A positional-indexing error in the per-class extraction utility misreported R_{normal} for single-class holdout folds (notably P13) through Phase 4; it is a measurement artefact, not a model error, and is documented in full in Measurement Notes. Affected folds are excluded from per-class means, and the underlying detections are correct.

False-positive propagation into the displayed ratio. Because the application reports the atypical fraction against the ≥25% threshold before the manual review step, the false-positive rate is not purely cosmetic: it feeds the numerator and denominator that the screen value is computed from. The error is also asymmetric. In the G3 confusion matrix (Figure 15), the true-background column is split 0.91 to predicted-atypical and 0.09 to predicted-normal, so a false positive arising from background tissue is labelled atypical about nine times out of ten. Each such artefact therefore inflates the numerator of the ratio more than the denominator, biasing the displayed fraction upward rather than leaving it unchanged. The magnitude of this effect is governed entirely by the operating confidence threshold, and it is the reason the threshold is not left at a default value. That confusion matrix is computed at Ultralytics' validation default of conf = 0.25; at a deployment threshold chosen from the P11 calibration cohort, where the out-of-sample histogram (Figure 22) separates the low-confidence false-positive bulk from the true-positive peak, the surviving false-positive count over a (in this case) ~12′000-tile slide is far smaller. The displayed fraction is thus an operating-point-dependent screening value subject to the haematologist's review, not a final ratio, and the residual false alarms that clear the threshold are exactly what the filtered review workflow is designed to surface. The direct metric for this risk is the absolute error between the model's predicted atypical fraction and the ground-truth fraction at the chosen operating point; reporting it on a dedicated calibration cohort is the cleanest way to quantify clinical usability and is identified as a deployment-evaluation step in Section 5.3..

This same false-positive behaviour limits precision on the SM-dominant folds in the per-patient and grouped analyses (P2, P4; groups G1, G2), where atypical recall stays high while precision falls. For P1 the precision shortfall is instead the annotation-regime effect described above, not a property of the model's false-positive behaviour.

Out-of-distribution generalization. The most important limitation is structural: the training corpus is drawn almost entirely from confirmed or suspected SM patients, so the model has learned the mast-cell morphology distribution of an SM-enriched population. Its behaviour on aspirates from patients with unrelated haematological conditions, or from healthy controls, is out of distribution and not characterized by this work. The reported metrics describe performance within the SM diagnostic workflow they were built for, and should not be read as evidence of general bone-marrow cell-classification performance.

5. Conclusion

5.1. Summary of Key Findings

This thesis delivers the first patient-held-out-validated deep learning pipeline for distinguishing atypical from normal mast cells in bone marrow aspirates and aligning the result to the WHO ≥25% minor criterion for systemic mastocytosis. A YOLO11n detector trained on a 53-patient corpus (35 with annotated cells) generalizes across patients under grouped cross-validation at mAP50-95 = 0.748, R_{atypical} = 0.885, and R_{normal} = 0.821, reaching R_{normal} = 0.981 on the strongest multi-patient fold, with a 35-fold LOPO run bounding the optimistic case at R_{atypical} = 0.920 and R_{normal} = 0.941. Three findings carry beyond the numbers. First, a bootstrapped model-assisted annotation loop, in which each model generation labels the next batch and its confirmed false positives become the next generation's hard negatives, made a 53-patient rare-cell corpus feasible within a single thesis. Second, a controlled design-of-experiment result showed that under structural class imbalance, corpus expansion improves rare-class recall more than any hyperparameter change, a transferable lesson for rare-event medical imaging. Third, the work ships a usable artefact: the MastCellDetector desktop application runs the model offline at USZ, reports the atypical fraction against the ≥25% threshold, and was accompanied by an out-of-sample confidence calibration so the hospital sets its own operating point.

5.2. Limitations and Challenges

The corpus is drawn almost entirely from SM-enriched patients, so performance on non-SM or healthy aspirates is out of distribution and uncharacterized. The minority normal class remains the harder direction, and per-class recall on small single-patient folds is dominated by sampling noise rather than model behaviour, which is why the conservative grouped-CV estimate, not the ceiling-inflated LOPO mean, is treated as primary.

A further limitation follows from the annotation method itself. Ground truth was produced through a bootstrapped, model-assisted CVAT loop in which each model generation proposed bounding boxes that the annotator then accepted, corrected, or rejected. This procedure is effective at removing false positives, since proposed boxes are reviewed explicitly, but it offers no equivalent mechanism for recovering false negatives: a genuine mast cell that the model failed to propose is one the annotator was also less likely to add, a form of automation bias documented in computer-assisted reading. The ground-truth corpus is therefore mildly enriched for cells the detector family already finds, and the reported recall is strictly recall against this model-assisted ground truth rather than against an independent exhaustive re-read. The first patient was annotated by hand without model assistance, which bounds but does not eliminate the effect. The most direct remedy is to quantify the bias: a blind, unassisted re-annotation of a random tile subset by an independent reader, compared against the model-assisted ground truth, would estimate the residual false-negative rate and bound the recall bias. The loop itself can also be hardened by decoupling the proposer from the evaluator, generating candidate boxes with an independently trained model or ensemble, and by lowering the proposal threshold so that more candidates enter explicit review and the annotator stays in the reject mode the loop is strong at, rather than having to recover missed cells unaided.

No second-institution external validation set was available, so LOPO and grouped patient CV are the strongest generalization estimates obtainable from a single site. The per-class recall measurement bug, although corrected for in the reported means, affected single-class folds during development. Finally, the operating confidence threshold is deliberately left to the clinic: the delivered calibration is a single near-balanced patient (P11), and a larger dedicated calibration set would tighten it.

5.3. Future Research Directions

Several directions follow naturally. The full annotated corpus can now train a final model, evaluated under the P5-A leave-one-patient-out protocol, whose near-complete training folds best approximate a model trained on the whole corpus; at the expanded scale the small-sample ceiling folds that made P5-A optimistic in this work largely disappear, so the estimate becomes both low-bias and representative. A dedicated, class-balanced held-out calibration set, larger than the single P11 patient, would give the hospital a cleaner out-of-sample threshold than one patient can support; reporting the absolute error between the predicted and ground-truth atypical fraction on that cohort would add the first direct measure of clinical screening accuracy. The clearest validity step is external validation on aspirates from a second institution, which would test the SM-enriched training distribution against genuinely independent staining and scanning. On the deployment side, integration into Cinderella, the cell-classification software already in use at USZ, becomes worthwhile once the institutional security and change-management timeline permits, turning the standalone tool into part of the routine scan-to-inference-to-evaluation pipeline.

6. Declaration on AI Assistance

The author declares that the empirical research, data analysis, experimental design, and quantitative results presented in this thesis were generated independently and strictly remain the intellectual property of the author.

Artificial intelligence (AI) assistance was utilized solely for the purpose of language refinement, stylistic editing, optimization of academic reading flow and coding assistance. Specifically, the AI models (Gemini 2.5 Pro, Claude Sonnet 4.5, and Claude Sonnet 4.6) were employed to translate text, correct grammar, and restructure sentences into the required third-person past tense and formal academic style.

No core scientific content, model selection decisions, hyperparameter choices, or interpretation of results were delegated to or generated by the AI model.

7. References

[1] P. Valent *et al.*, “Updated diagnostic criteria and classification of mast cell disorders: A consensus proposal,” *HemaSphere*, vol. 5, no. 11, p. e646, Oct. 2021, doi: 10.1097/HS9.0000000000000646.

[2] K. Y. Tse *et al.*, “MASTering systemic mastocytosis: Lessons learned from a large patient cohort,” *Journal of Allergy and Clinical Immunology: Global*, vol. 3, no. 4, p. 100316, 2024, doi: https://doi.org/10.1016/j.jacig.2024.100316.

[3] C. Ustun, F. Keklik Karadag, M. A. Linden, P. Valent, and C. Akin, “Systemic mastocytosis: Current status and challenges in 2024,” *Blood Advances*, vol. 9, no. 9, pp. 2048–2062, Apr. 2025, doi: 10.1182/bloodadvances.2024012612.

[4] S. Choudhary, S. Kumar, P. Siddhaarth, *et al.*, “Advancing blood cell detection and classification: Performance evaluation of modern deep learning models,” *BMC Medical Informatics and Decision Making*, vol. 25, no. 207, 2025, doi: 10.1186/s12911-025-03027-2.

[5] H. Sazak and M. Kotan, “Automated blood cell detection and classification in microscopic images using YOLOv11 and optimized weights,” *Diagnostics*, vol. 15, no. 1, 2025, doi: 10.3390/diagnostics15010022.

[6] M. L. Ali and Z. Zhang, “The YOLO framework: A comprehensive review of evolution, applications, and benchmarks in object detection,” *Computers*, vol. 13, no. 12, 2024, doi: 10.3390/computers13120336.

[7] I. N. Margret and K. Rajakumar, “Deep learning techniques for analyzing peripheral blood smears: A meta-analysis,” *Neural Computing and Applications*, vol. 37, pp. 18039–18065, 2025, doi: 10.1007/s00521-025-11401-4.

[8] S. Glüge, S. Balabanov, V. H. Koelzer, and T. Ott, “Evaluation of deep learning training strategies for the classification of bone marrow cell images,” *Computer Methods and Programs in Biomedicine*, vol. 243, p. 107924, 2024, doi: https://doi.org/10.1016/j.cmpb.2023.107924.

[9] T. Ghete *et al.*, “Models for the marrow: A comprehensive review of AI-based cell classification methods and malignancy detection in bone marrow aspirate smears,” *HemaSphere*, vol. 8, no. 12, p. e70048, 2024, doi: https://doi.org/10.1002/hem3.70048.

[10] J. Diaz, A. Deb, A. Lyubimova, L. Santos-Carrion, C. Romagnoli, and J. Barrientos, “Machine learning–based detection model for leukemia cells in peripheral blood smears using YOLOv11-large,” *Blood*, vol. 146, p. 6123, 2025, doi: https://doi.org/10.1182/blood-2025-6123.

[11] Z. Lv, X. Cao, X. Jin, *et al.*, “High-accuracy morphological identification of bone marrow cells using deep learning-based morphogo system,” *Scientific Reports*, vol. 13, no. 13364, 2023, doi: 10.1038/s41598-023-40424-x.

[12] V. Srilakshmi, K. S. Chakradhar, K. Suneetha, C. S. Bindu, N. Yamsani, and K. R. Madhavi, "Artificial intelligence for detecting prevalence of indolent mastocytosis," in *Proceedings of the 14th international conference on soft computing and pattern recognition (SoCPaR 2022)*, A. Abraham, T. Hanne, N. Gandhi, P. Manghirmalani Mishra, A. Bajaj, and P. Siarry, Eds., Cham: Springer Nature Switzerland, 2023, pp. 33–43.

[13] F. Vollmer, "GitHub Repository: Glass Defect Detection — Evaluating Object Models." Dec. 2025. Available: https://github.zhaw.ch/vollmflo/Glass-Defect-Detection-Evaluating-Object-Detection-Models

[14] K. Amin, "The role of mast cells in allergic inflammation," *Respiratory Medicine*, vol. 106, no. 1, pp. 9–14, Jan. 2012, doi: 10.1016/j.rmed.2011.09.007.

[15] Cleveland Clinic, "Mast cells: Anatomy, function & diseases." Accessed: Apr. 14, 2026. [Online]. Available: https://my.clevelandclinic.org/health/body/mast-cells

[16] T. C. Theoharides *et al.*, "Mast cells and inflammation," *Biochimica et Biophysica Acta (BBA) - Molecular Basis of Disease*, vol. 1822, no. 1, pp. 21–33, 2012, doi: https://doi.org/10.1016/j.bbadis.2010.12.014.

[17] R. Alyamany *et al.*, "Unraveling the rare entity of KIT D816V-negative systemic mastocytosis," *Journal of Hematology*, vol. 13, no. 3, 2024, Available: https://www.thejh.org/index.php/jh/article/view/1279

[18] J. Y. Li *et al.*, "Review and updates on systemic mastocytosis and related entities," *Cancers*, vol. 15, no. 23, 2023, doi: 10.3390/cancers15235626.

[19] A. C. Garcia-Montero *et al.*, "KIT D816V–mutated bone marrow mesenchymal stem cells in indolent systemic mastocytosis are associated with disease progression," *Blood*, vol. 127, no. 6, pp. 761–768, Feb. 2016, doi: 10.1182/blood-2015-07-655100.

[20] M. Jawhar *et al.*, "KIT D816 mutated/CBF-negative acute myeloid leukemia: A poor-risk subtype associated with systemic mastocytosis," *Leukemia*, vol. 33, no. 5, pp. 1124–1134, May 2019, doi: 10.1038/s41375-018-0346-z.

[21] C. Teodosio *et al.*, "An immature immunophenotype of bone marrow mast cells predicts for multilineage D816V KIT mutation in systemic mastocytosis," *Leukemia*, vol. 26, no. 5, pp. 951–958, May 2012, doi: 10.1038/leu.2011.293.

[22] B. Sekachev *et al.*, *Opencv/cvat: v1.1.0.* (Aug. 2020). Zenodo. doi: 10.5281/zenodo.4009388.

[23] A. V. Buslaev, A. Parinov, E. Khvedchenya, V. I. Iglovikov, and A. A. Kalinin, "Albumentations: Fast and flexible image augmentations," *CoRR*, vol. abs/1809.06839, 2018, Available: http://arxiv.org/abs/1809.06839

[24] G. Jocher and J. Qiu, *Ultralytics YOLO11.* (2024). Available: https://github.com/ultralytics/ultralytics

[25] G. Jocher, J. Qiu, and A. Chaurasia, *Ultralytics YOLO*. (Jan. 2023). Available: https://github.com/ultralytics/ultralytics

[26] R. Padilla, S. L. Netto, and E. A. B. da Silva, “A survey on performance metrics for object-detection algorithms,” in *2020 international conference on systems, signals and image processing (IWSSIP)*, 2020, pp. 237–242. doi: 10.1109/IWSSIP48289.2020.9145130.

[27] A. Abid, A. Abdalla, A. Abid, D. Khan, A. Alfozan, and J. Zou, *Gradio: Hassle-free sharing and testing of ML models in the wild*. (June 2019). doi: 10.48550/arXiv.1906.02569.

[28] F. Vollmer, “MastCellDetector: Desktop application for mast cell detection and classification.” 2026. Available: https://github.com/kekslol24/mastcelldetector

[29] F. Vollmer, “Automated detection and morphological classification of mast cells in bone marrow aspirates using deep learning.” GitHub, 2026. Available: https://github.com/kekslol24/BA

[22] Gemini 2.5 Pro. (n.d.). Google Gemini. Retrieved October 1, 2025, from https://gemini.google.com/app

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